

BFS Traversal

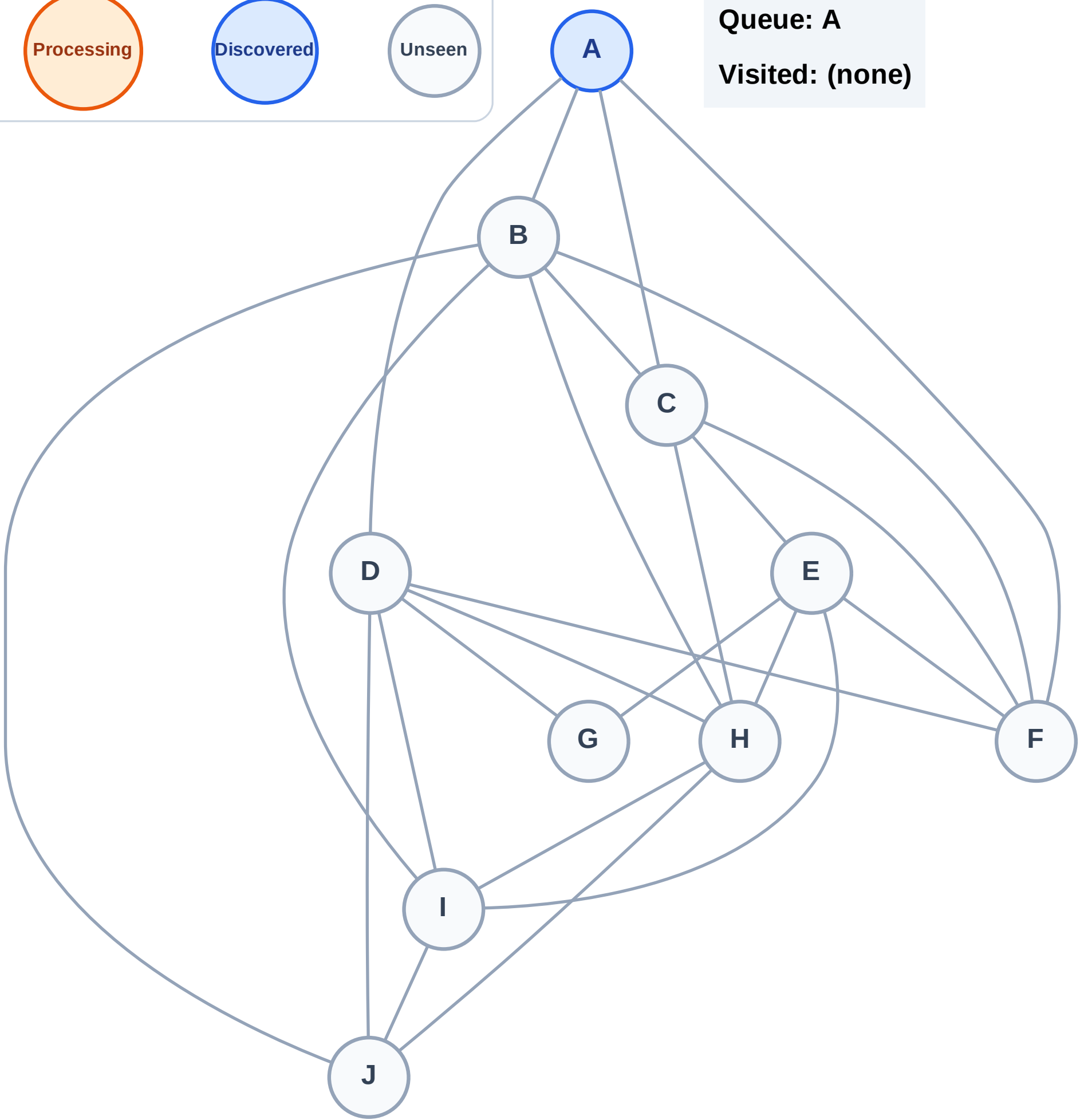
Step 1: Start at A

Legend



Queue: A

Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

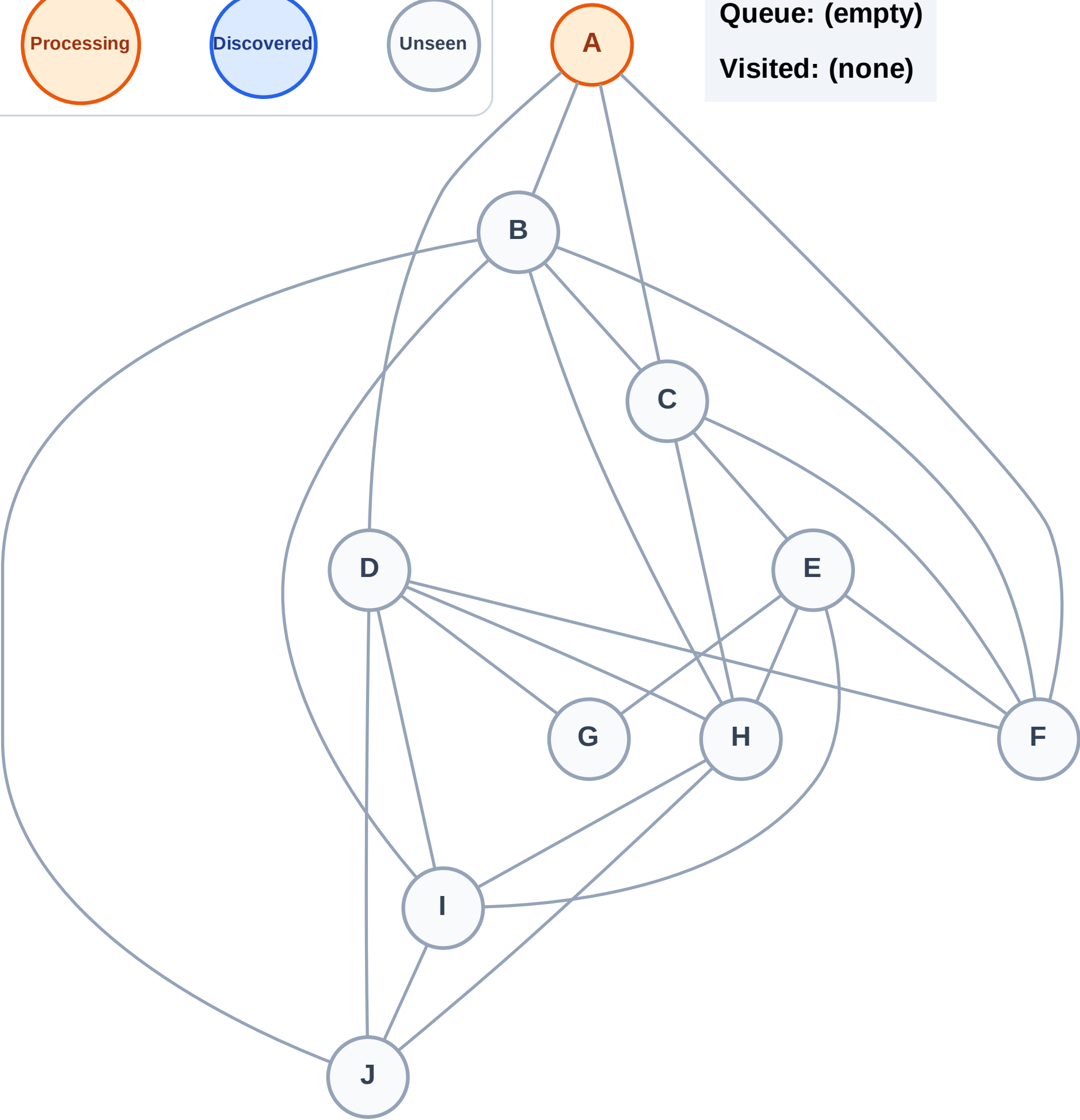
Complete

Processing

Discovered

Unseen

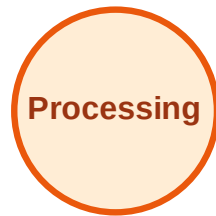
Queue: (empty)
Visited: (none)



BFS Traversal

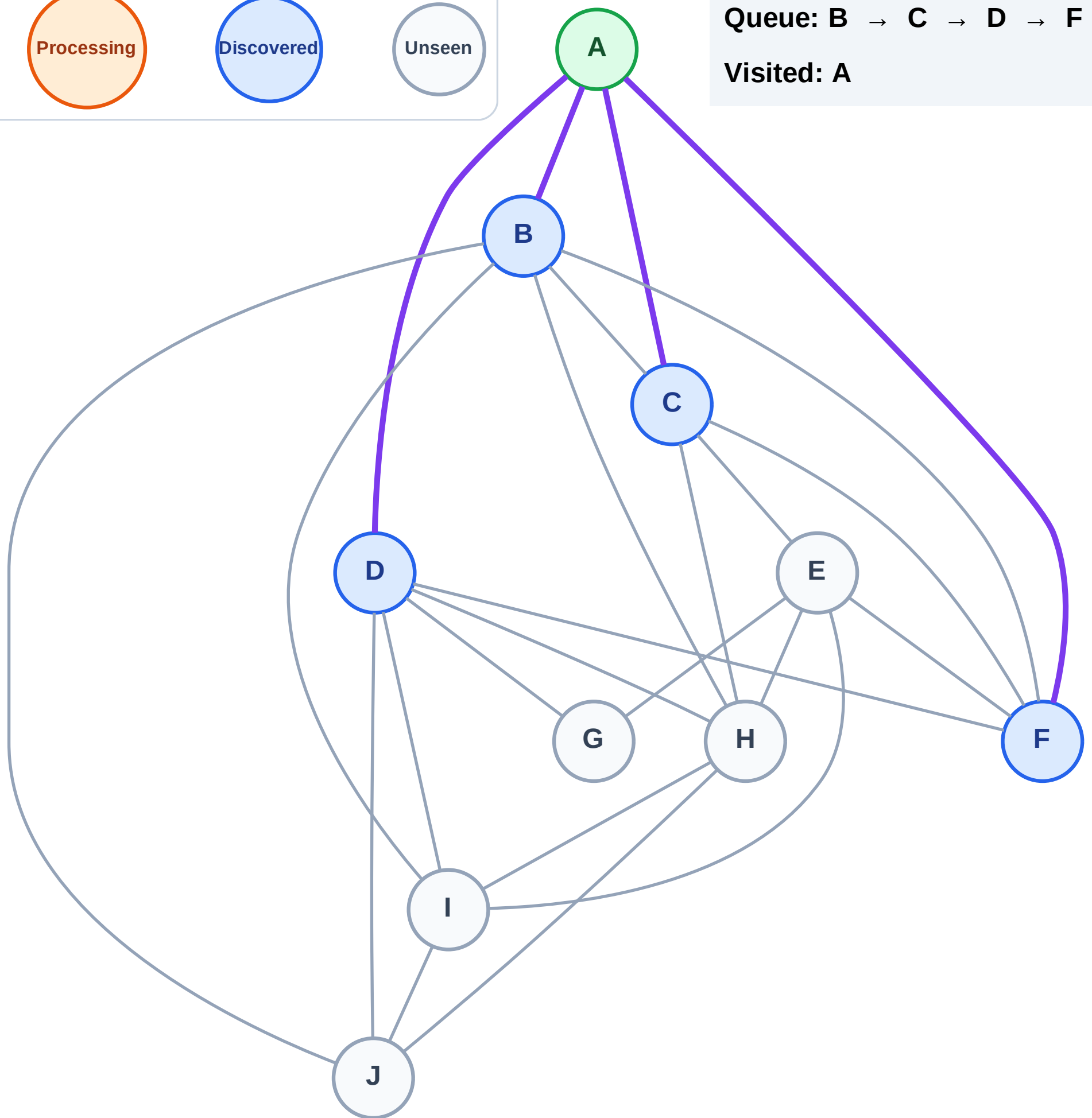
Step 3: Discover B, C, D, F from A

Legend



Queue: B → C → D → F

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

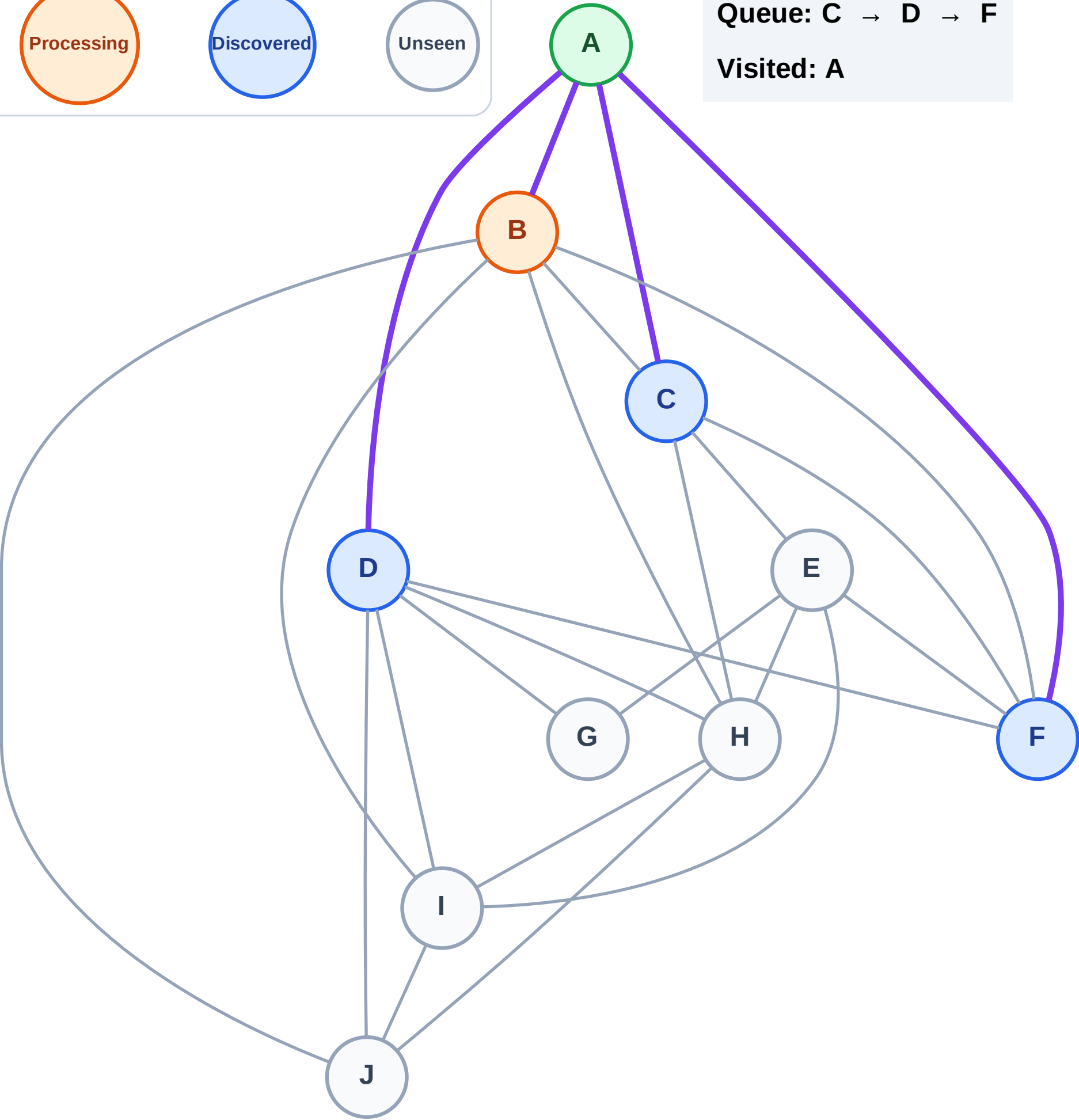
Processing

Discovered

Unseen

Queue: C → D → F

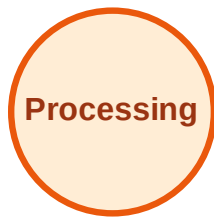
Visited: A



BFS Traversal

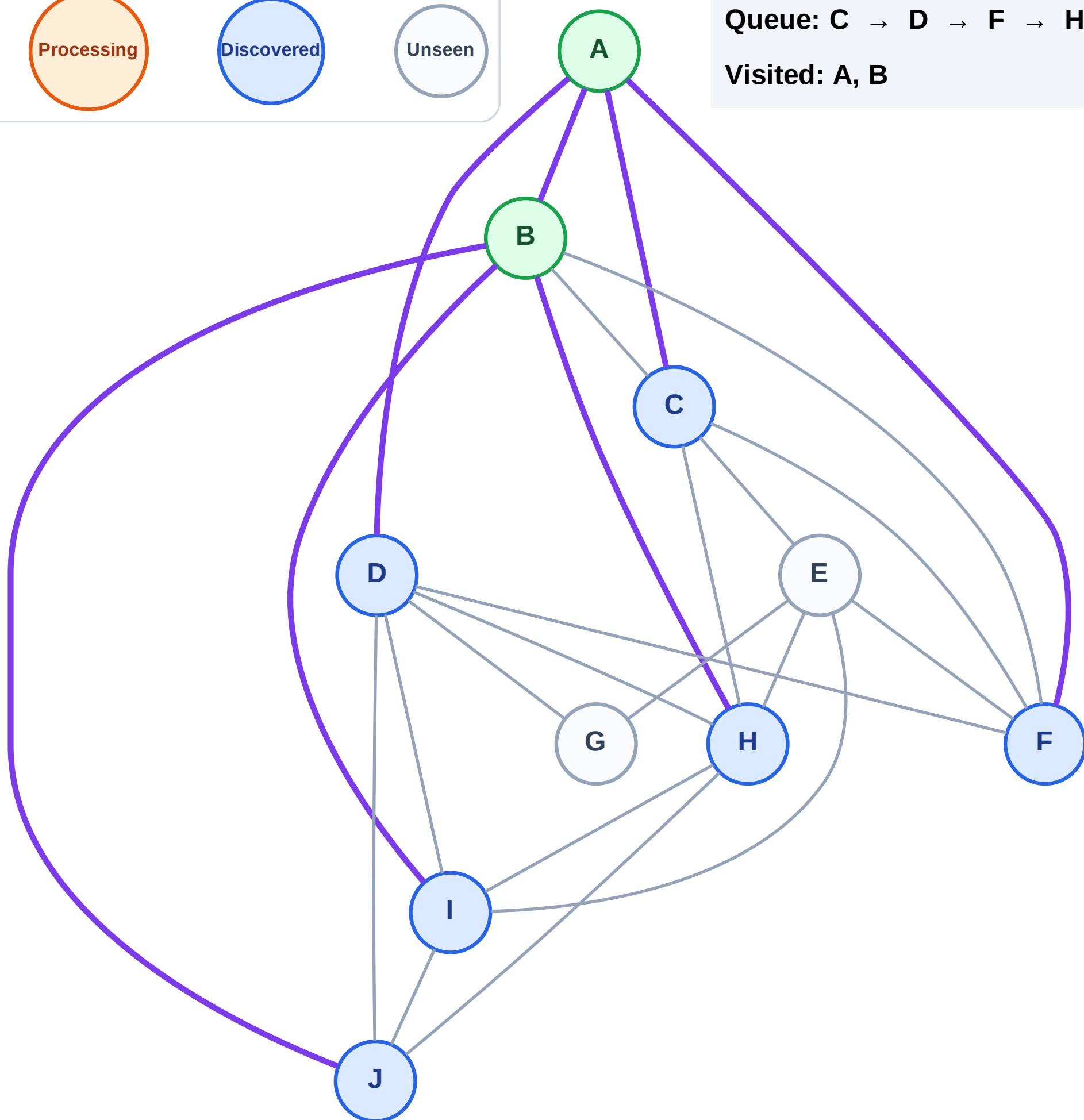
Step 5: Discover H, I, J from B

Legend



Queue: C → D → F → H → I → J

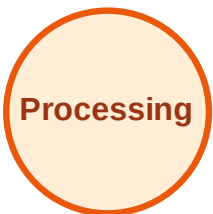
Visited: A, B



BFS Traversal

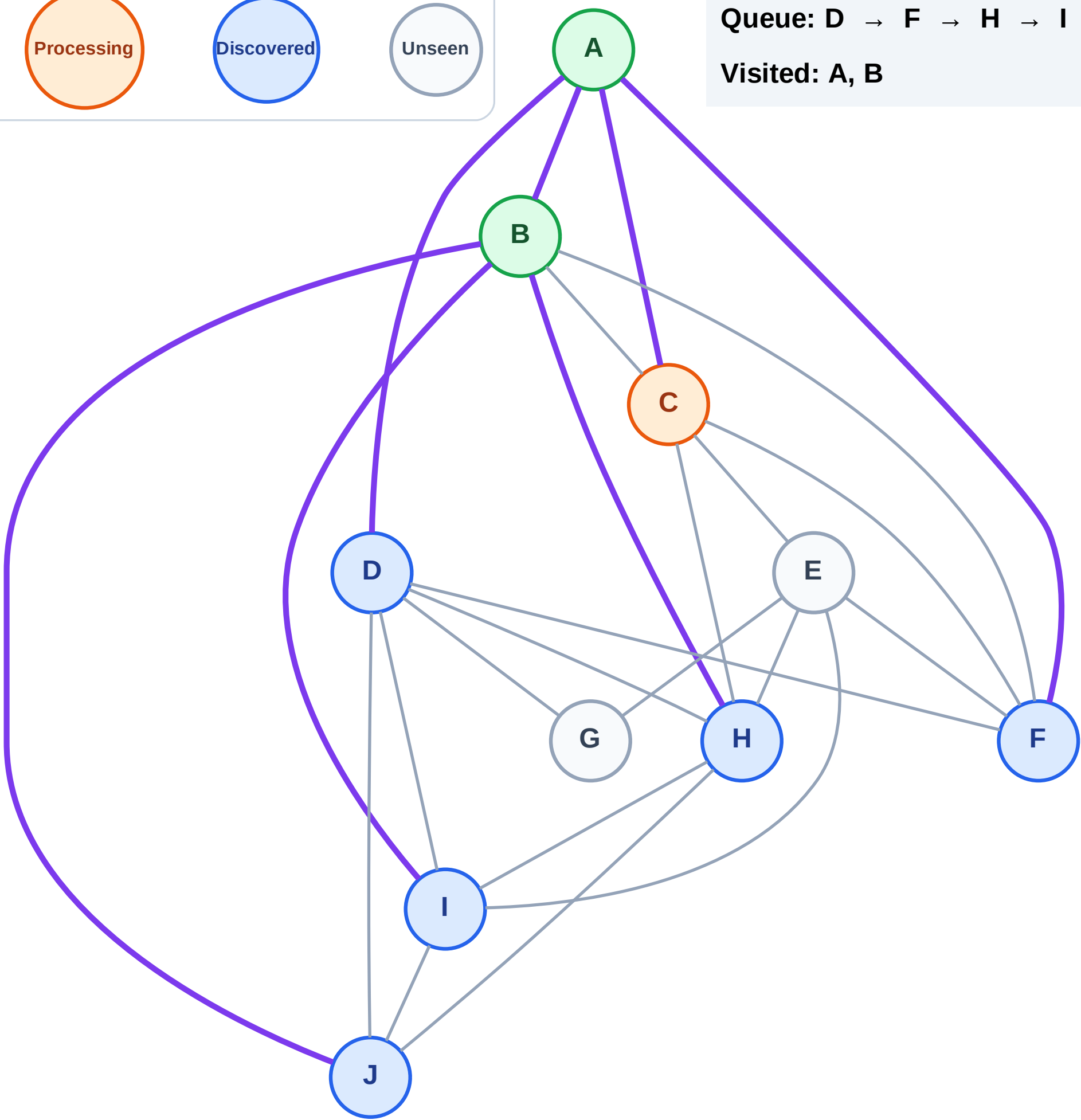
Step 6: Process node C

Legend



Queue: D → F → H → I → J

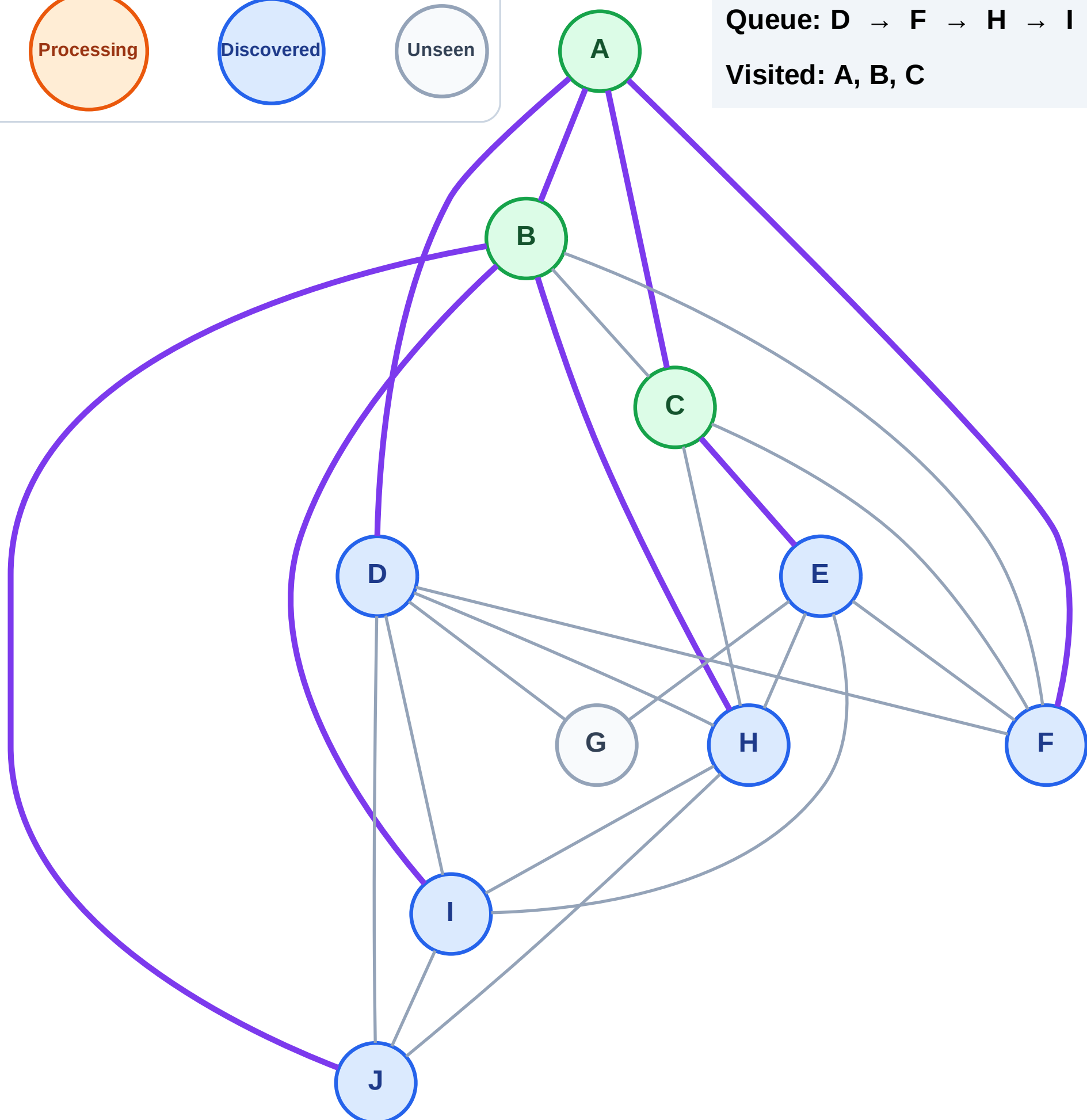
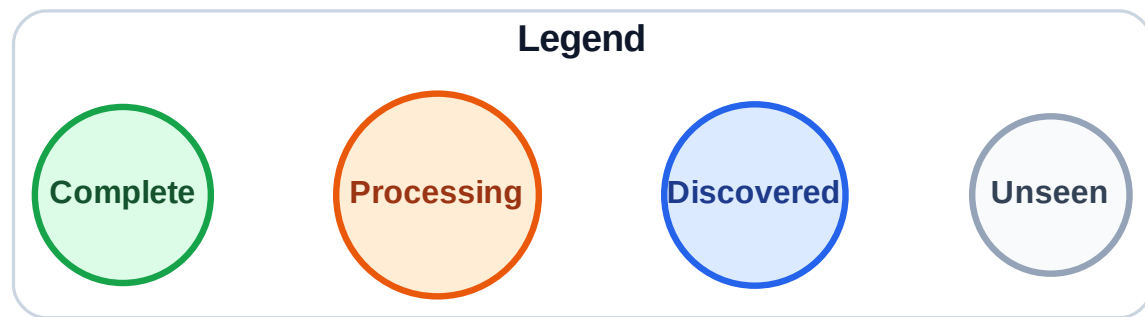
Visited: A, B



BFS Traversal

Step 7: Discover E from C

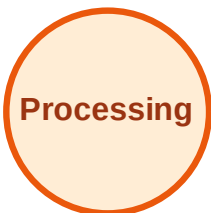
Queue: D → F → H → I → J → E
Visited: A, B, C



BFS Traversal

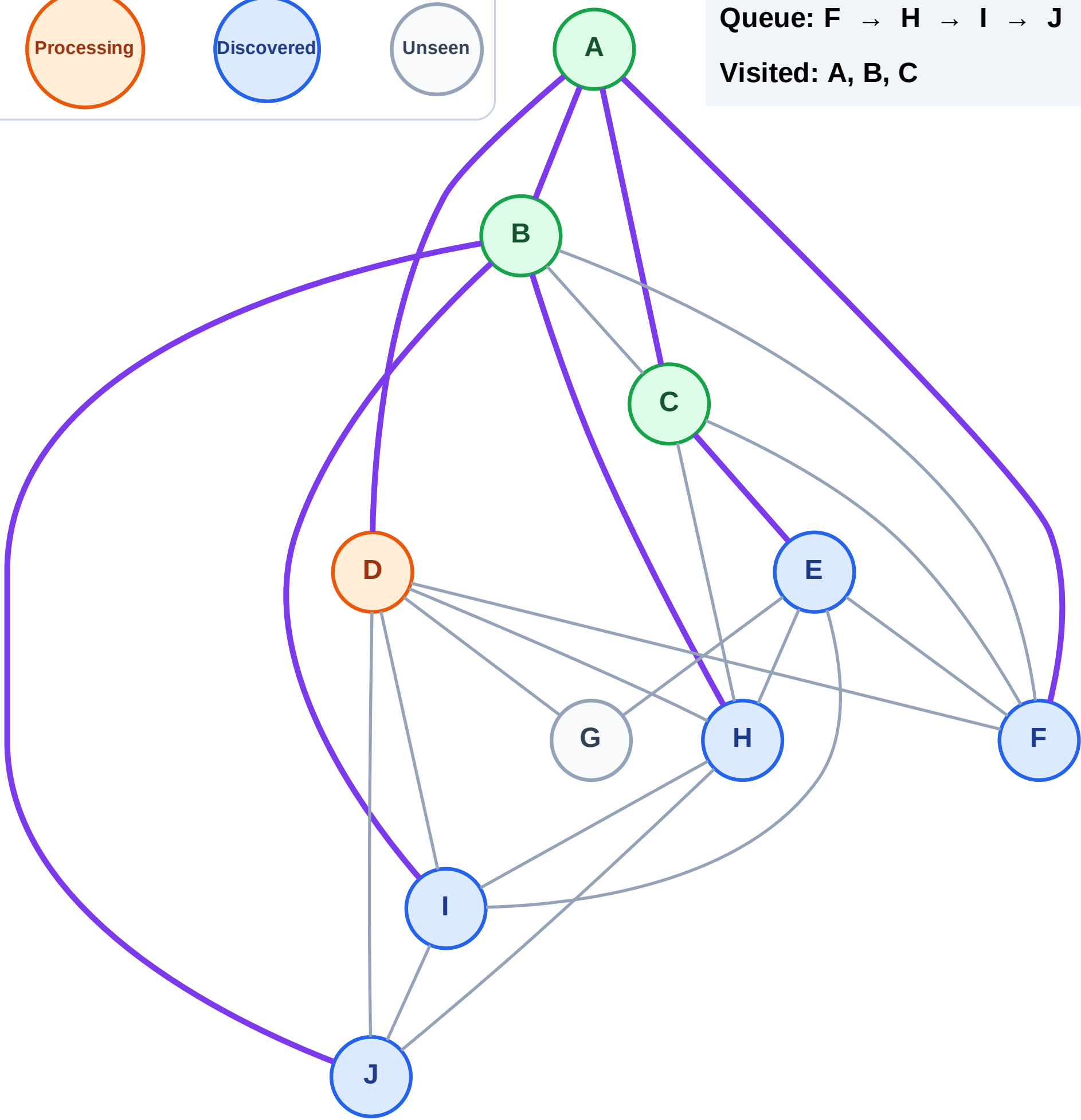
Step 8: Process node D

Legend



Queue: F → H → I → J → E

Visited: A, B, C

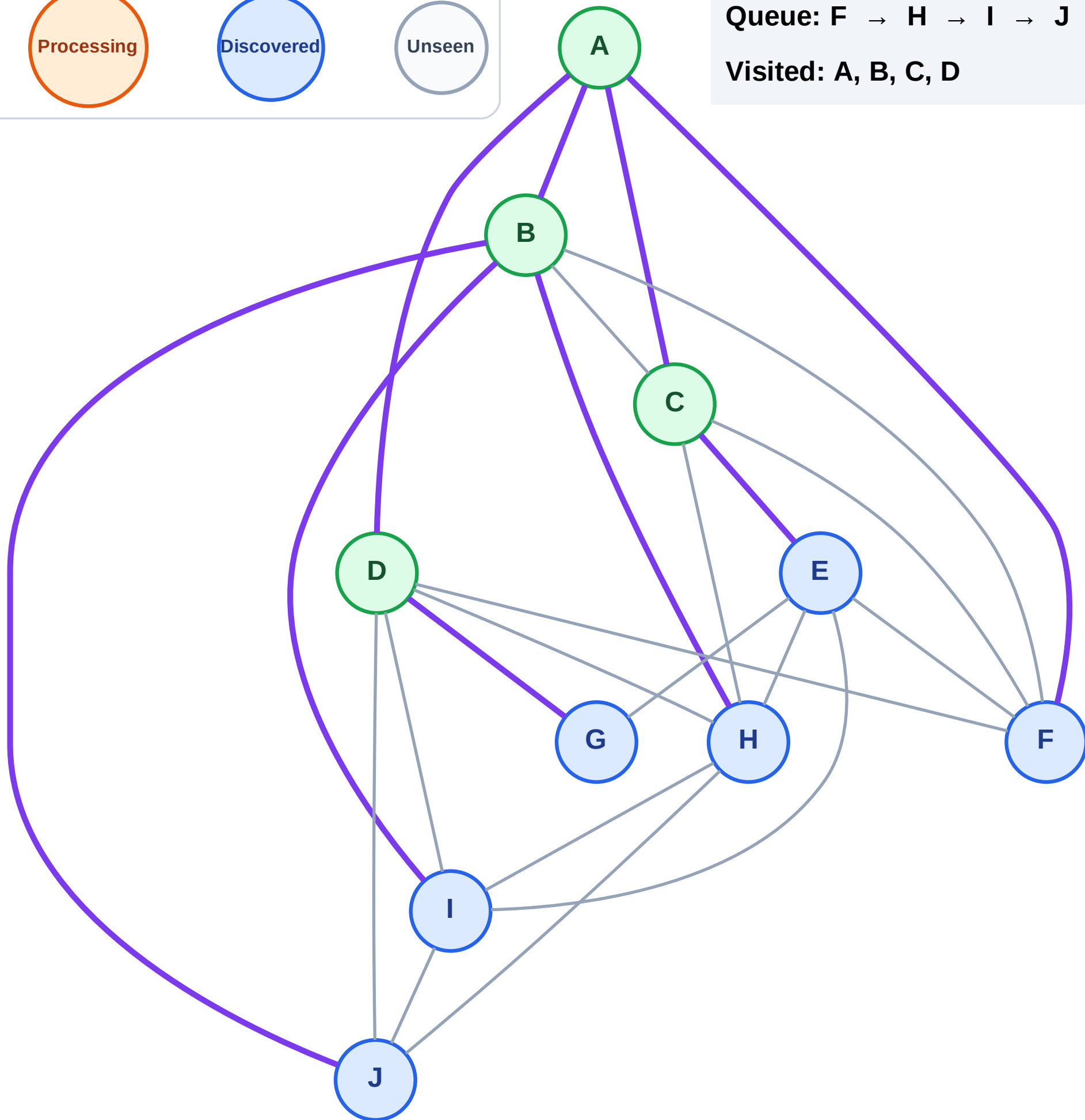
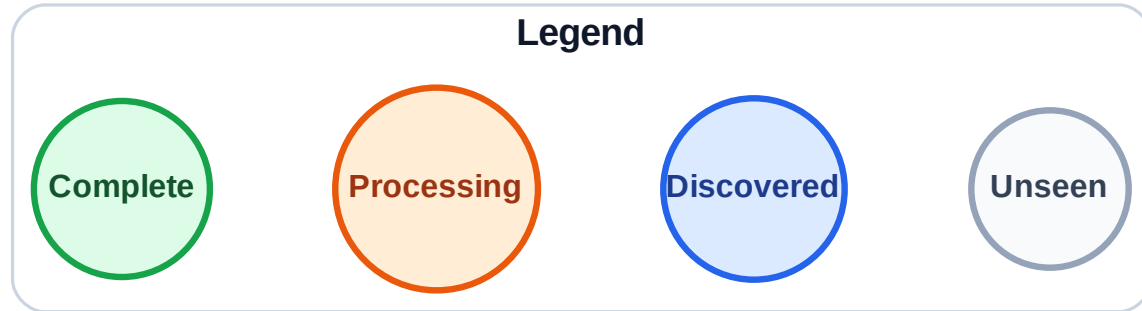


BFS Traversal

Step 9: Discover G from D

Queue: F → H → I → J → E → G

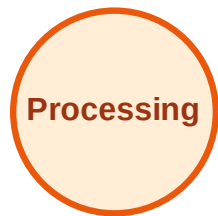
Visited: A, B, C, D



BFS Traversal

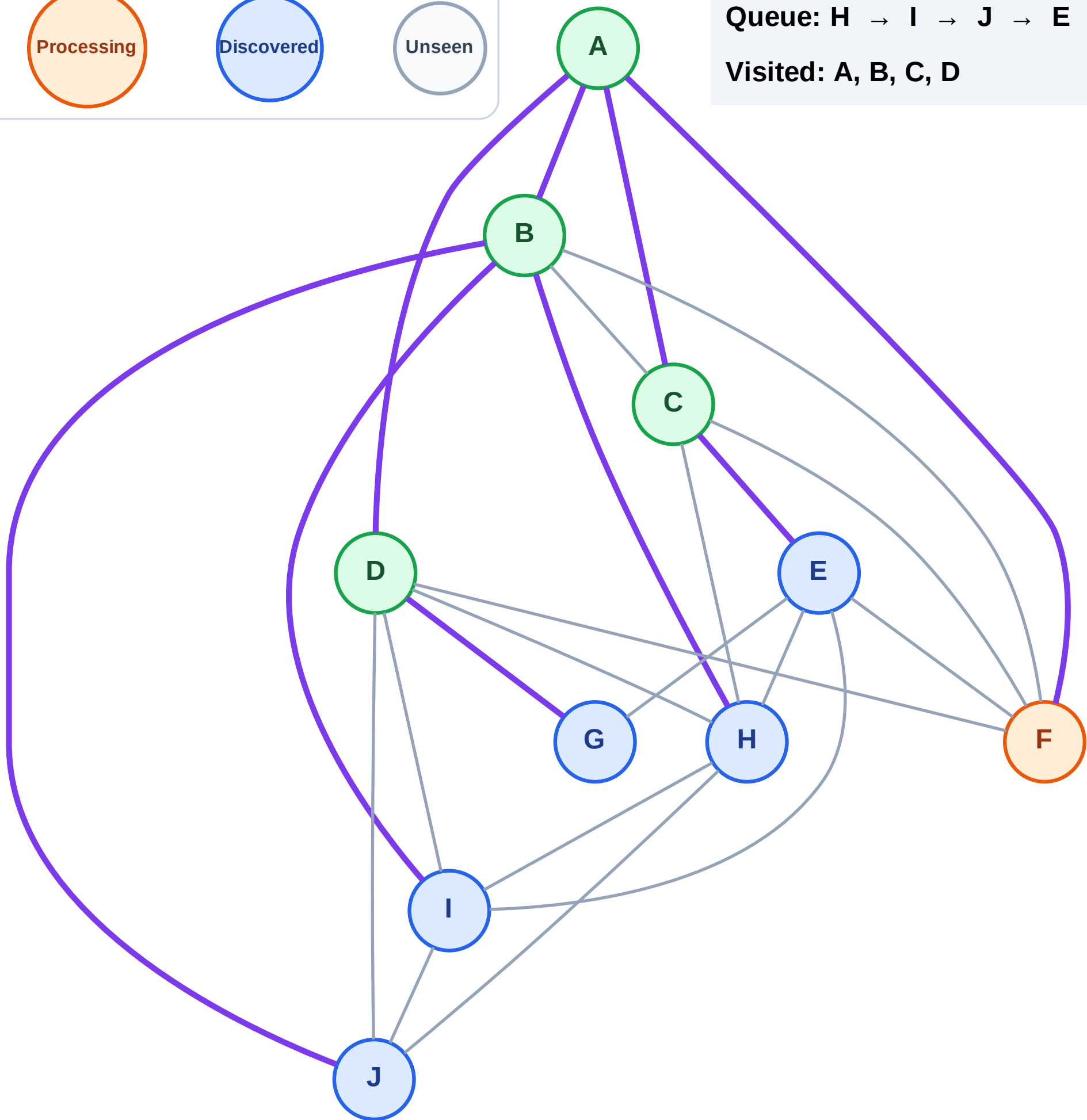
Step 10: Process node F

Legend



Queue: H → I → J → E → G

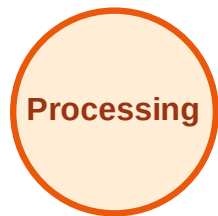
Visited: A, B, C, D



BFS Traversal

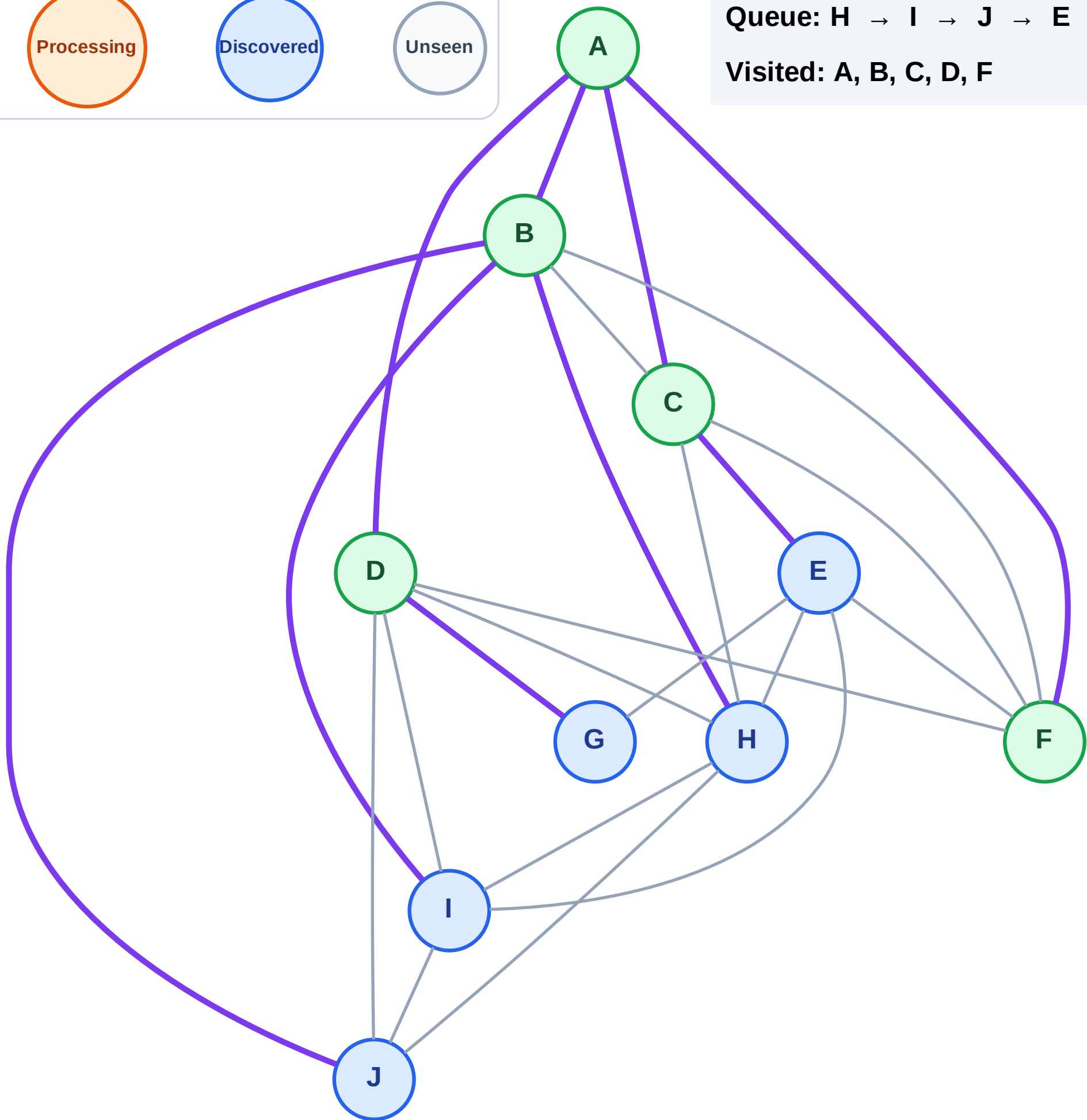
Step 11: No new neighbors from F

Legend



Queue: H → I → J → E → G

Visited: A, B, C, D, F



BFS Traversal

Step 12: Process node H

Legend

Complete

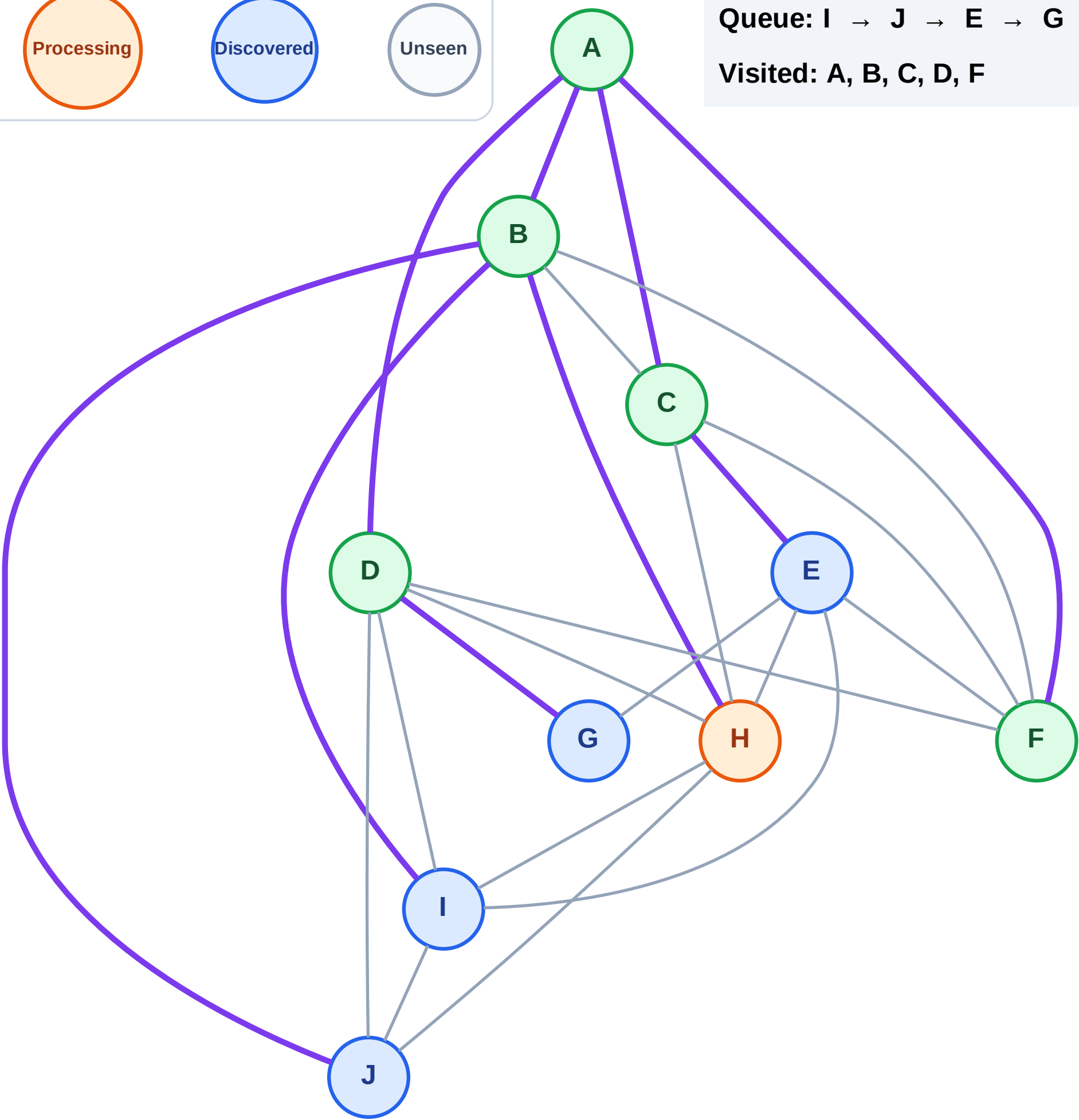
Processing

Discovered

Unseen

Queue: I → J → E → G

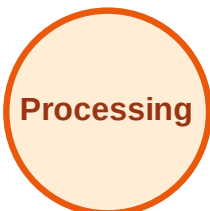
Visited: A, B, C, D, F



BFS Traversal

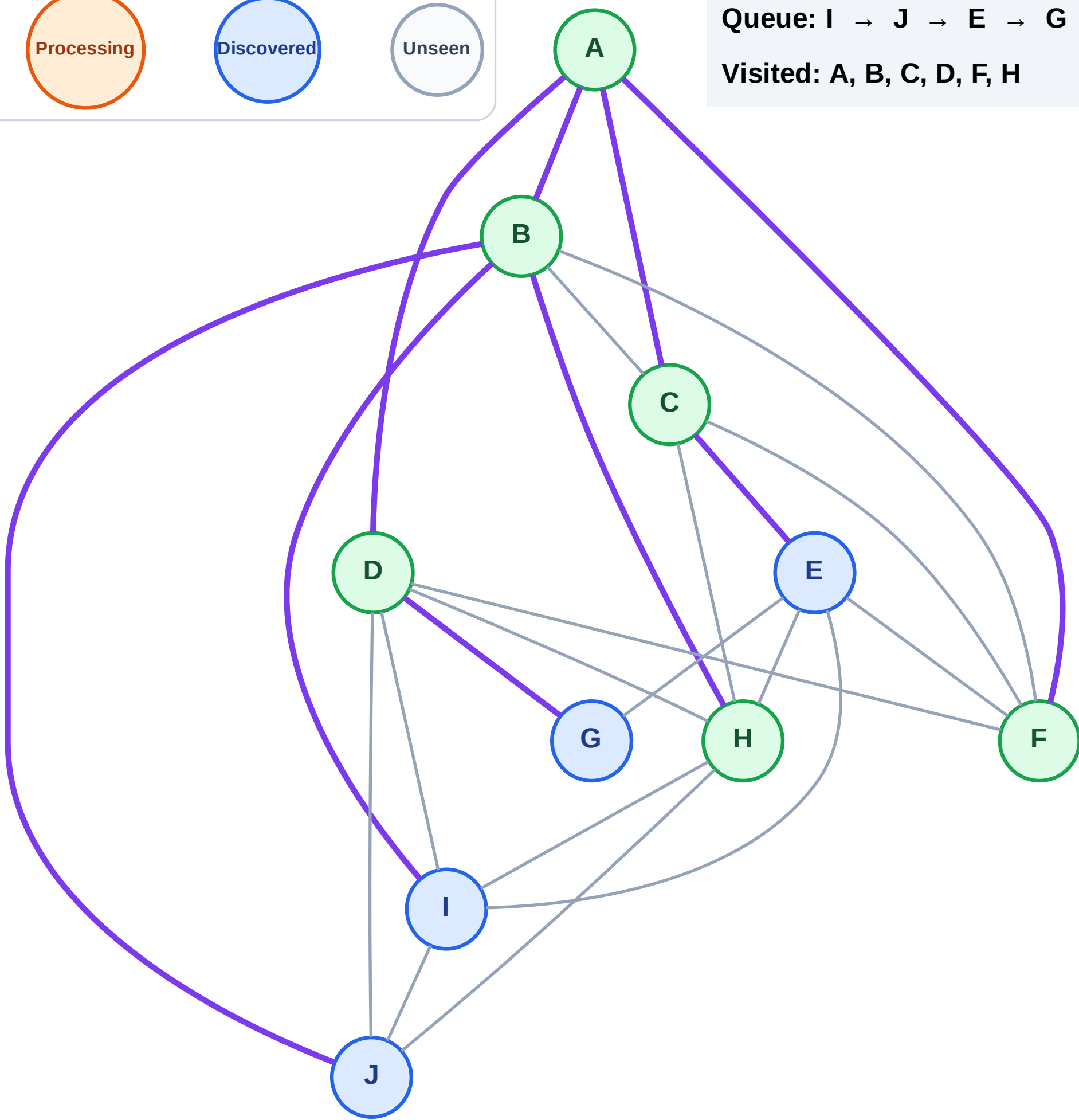
Step 13: No new neighbors from H

Legend



Queue: I → J → E → G

Visited: A, B, C, D, F, H



BFS Traversal

Step 14: Process node I

Legend

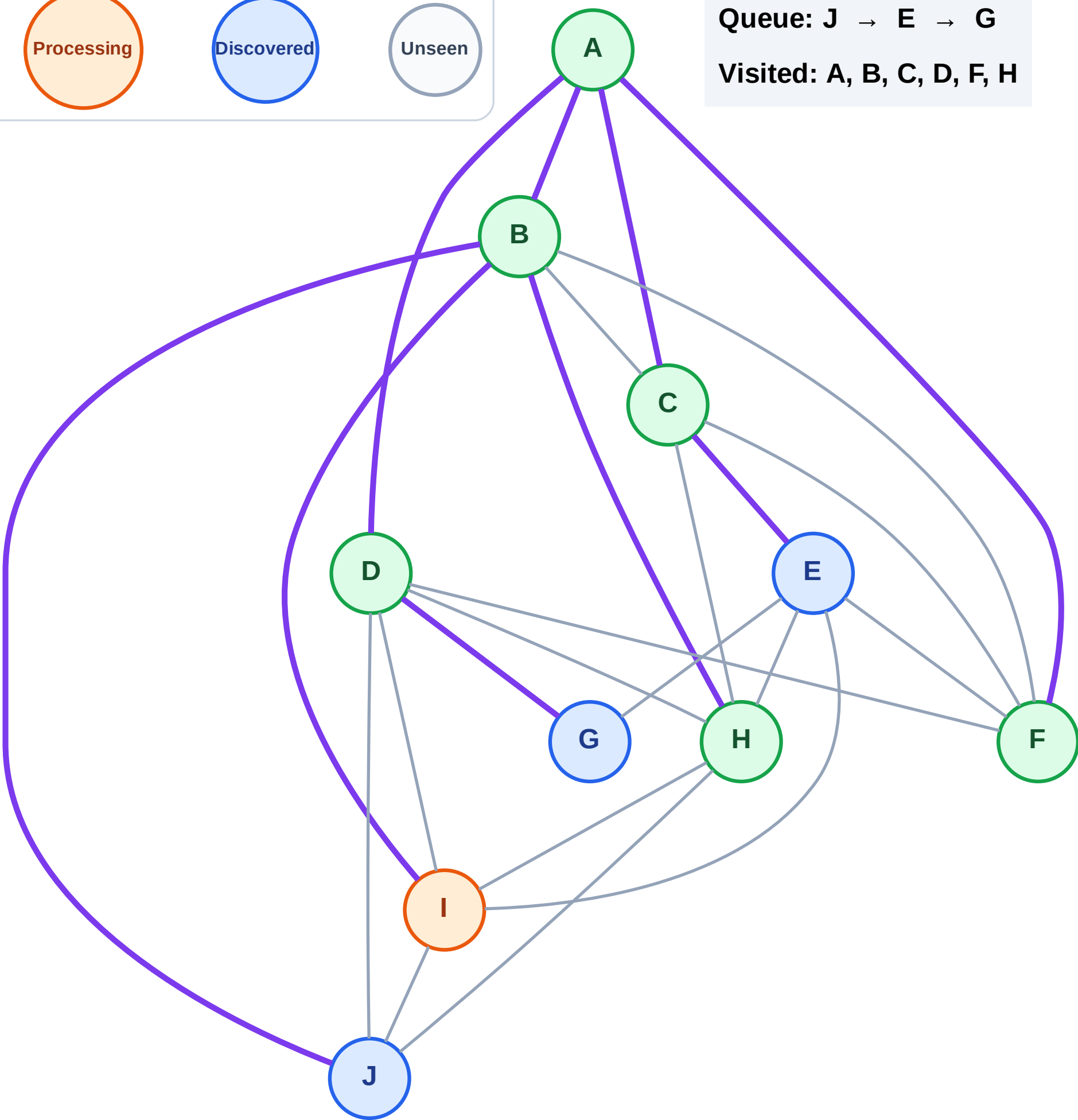
Complete

Processing

Discovered

Unseen

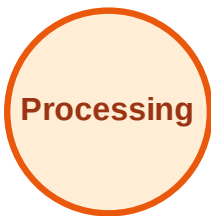
Queue: J → E → G
Visited: A, B, C, D, F, H



BFS Traversal

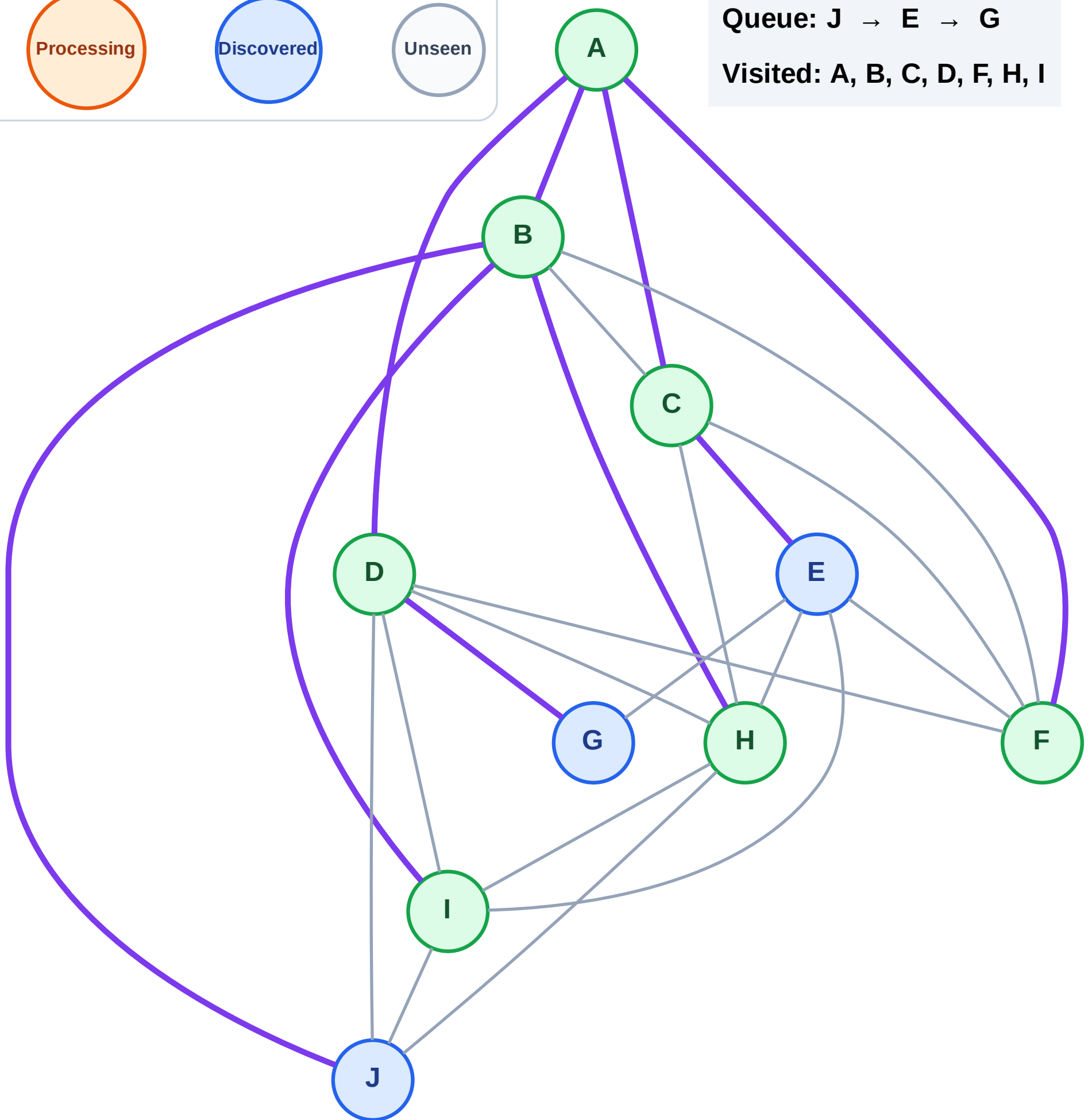
Step 15: No new neighbors from I

Legend



Queue: J → E → G

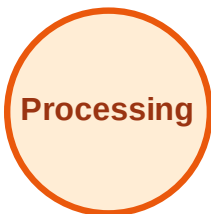
Visited: A, B, C, D, F, H, I



BFS Traversal

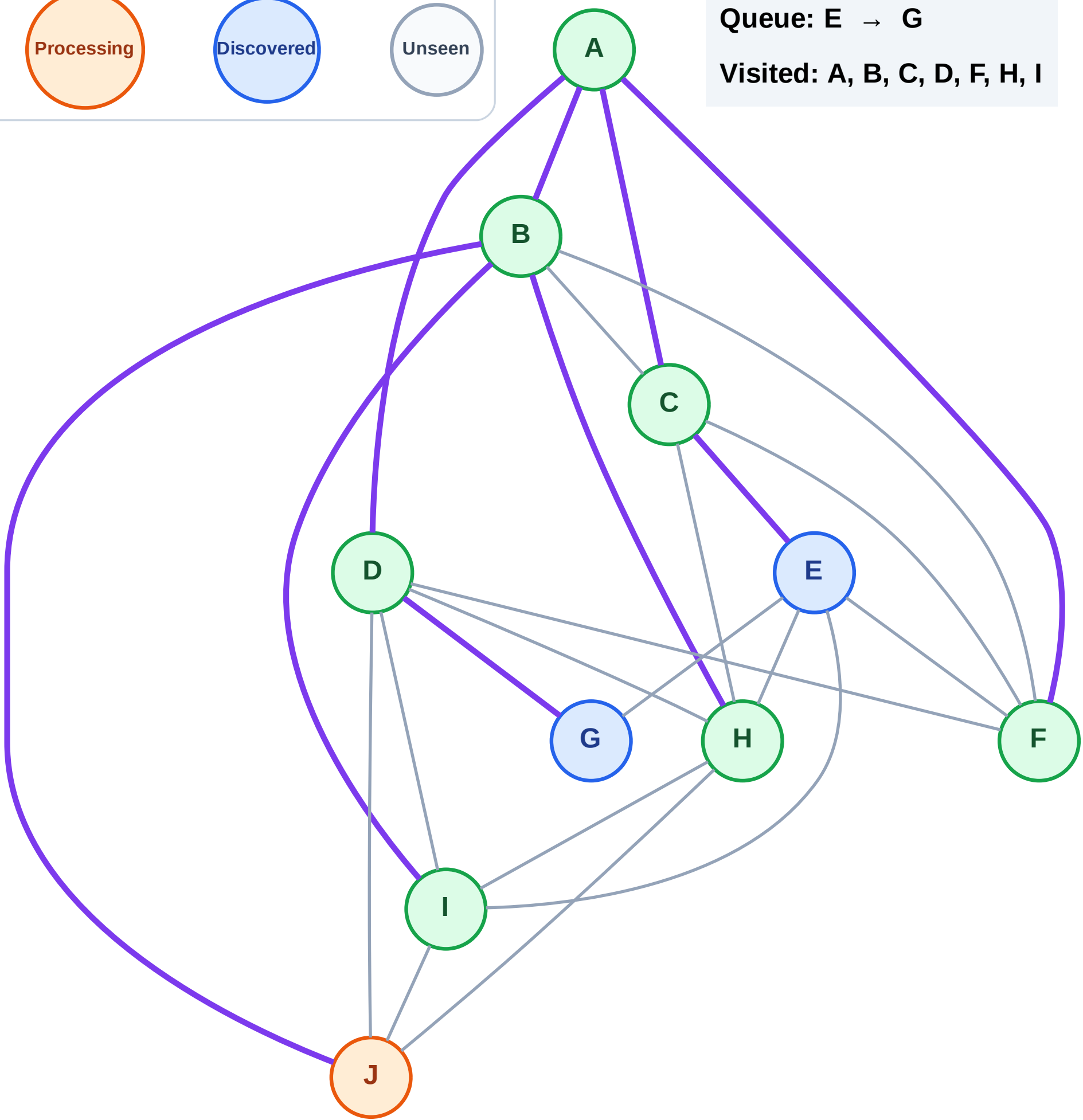
Step 16: Process node J

Legend



Queue: E → G

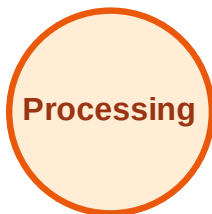
Visited: A, B, C, D, F, H, I



BFS Traversal

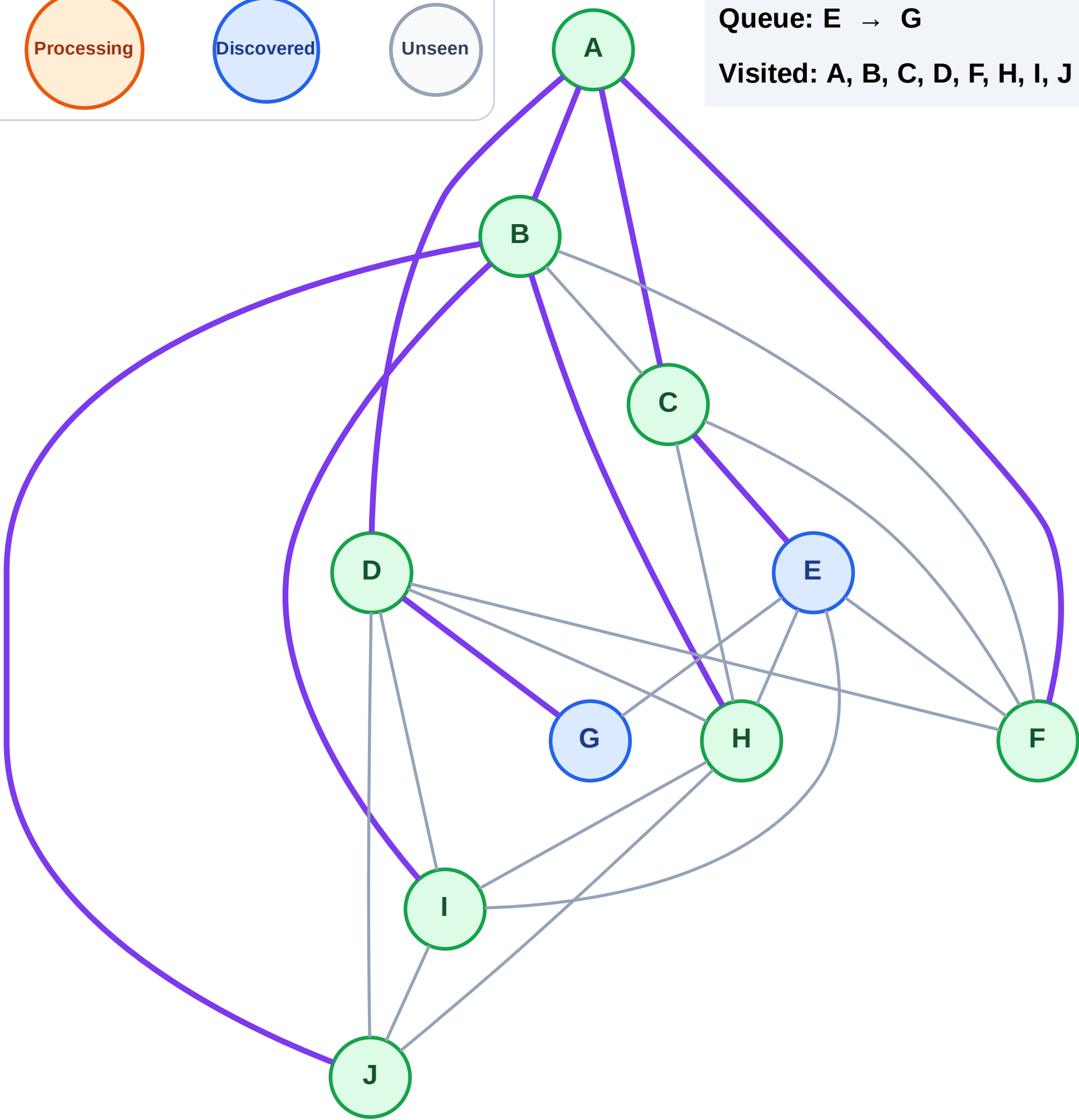
Step 17: No new neighbors from J

Legend



Queue: E → G

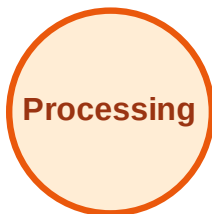
Visited: A, B, C, D, F, H, I, J



BFS Traversal

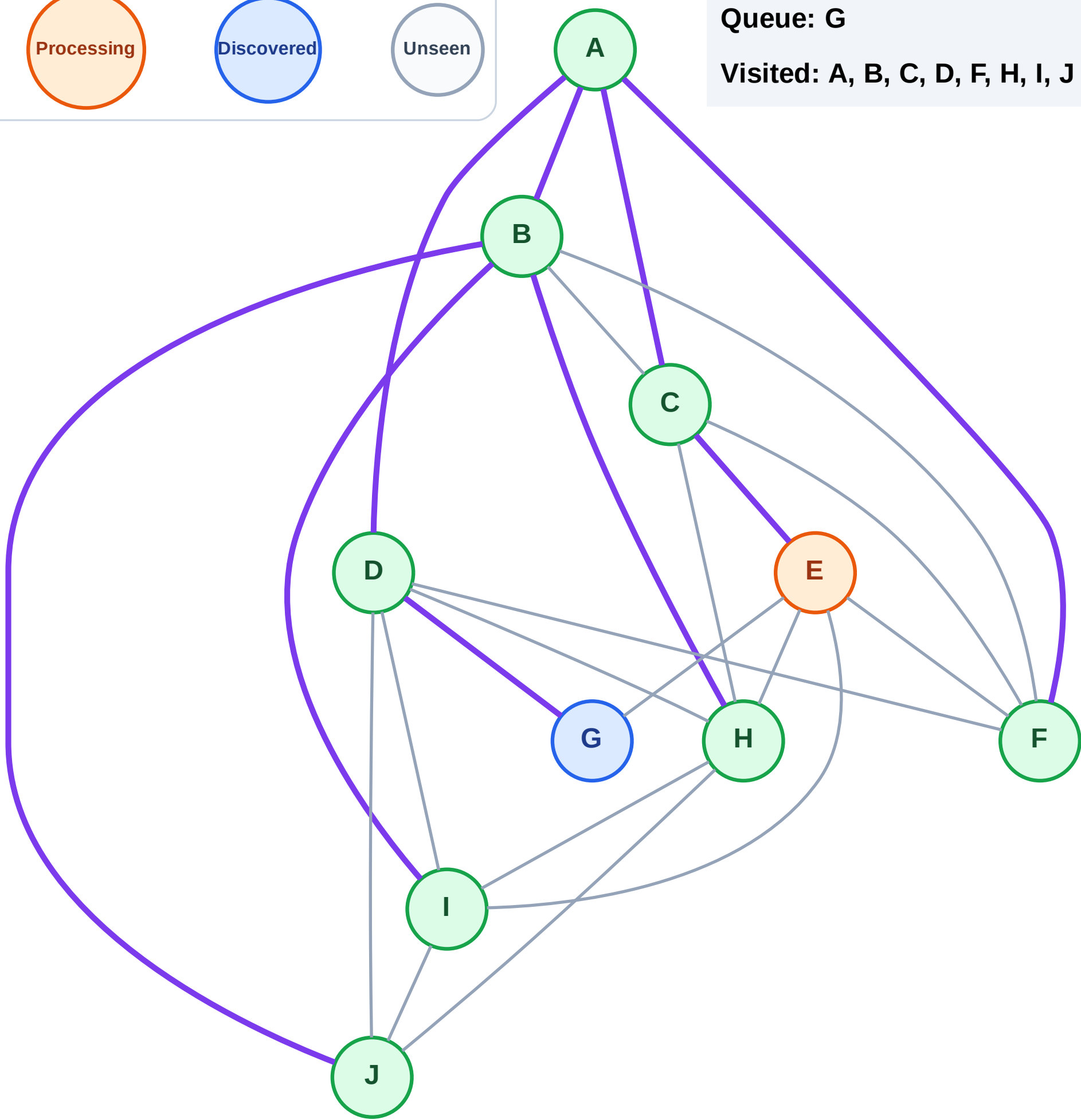
Step 18: Process node E

Legend



Queue: G

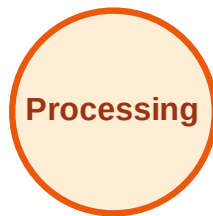
Visited: A, B, C, D, F, H, I, J



BFS Traversal

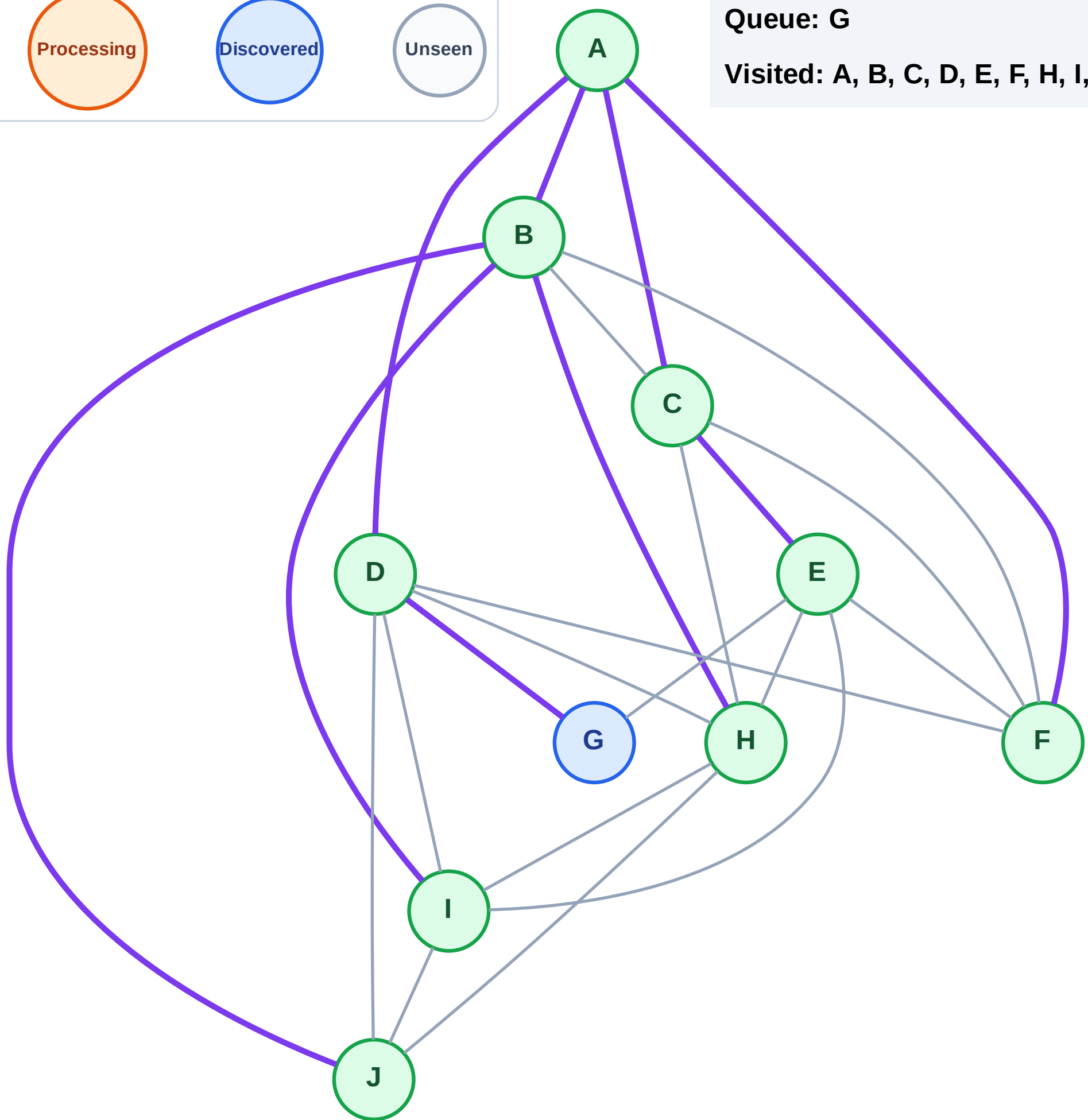
Step 19: No new neighbors from E

Legend



Queue: G

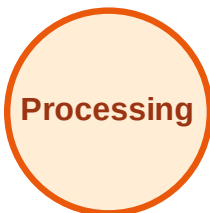
Visited: A, B, C, D, E, F, H, I, J



BFS Traversal

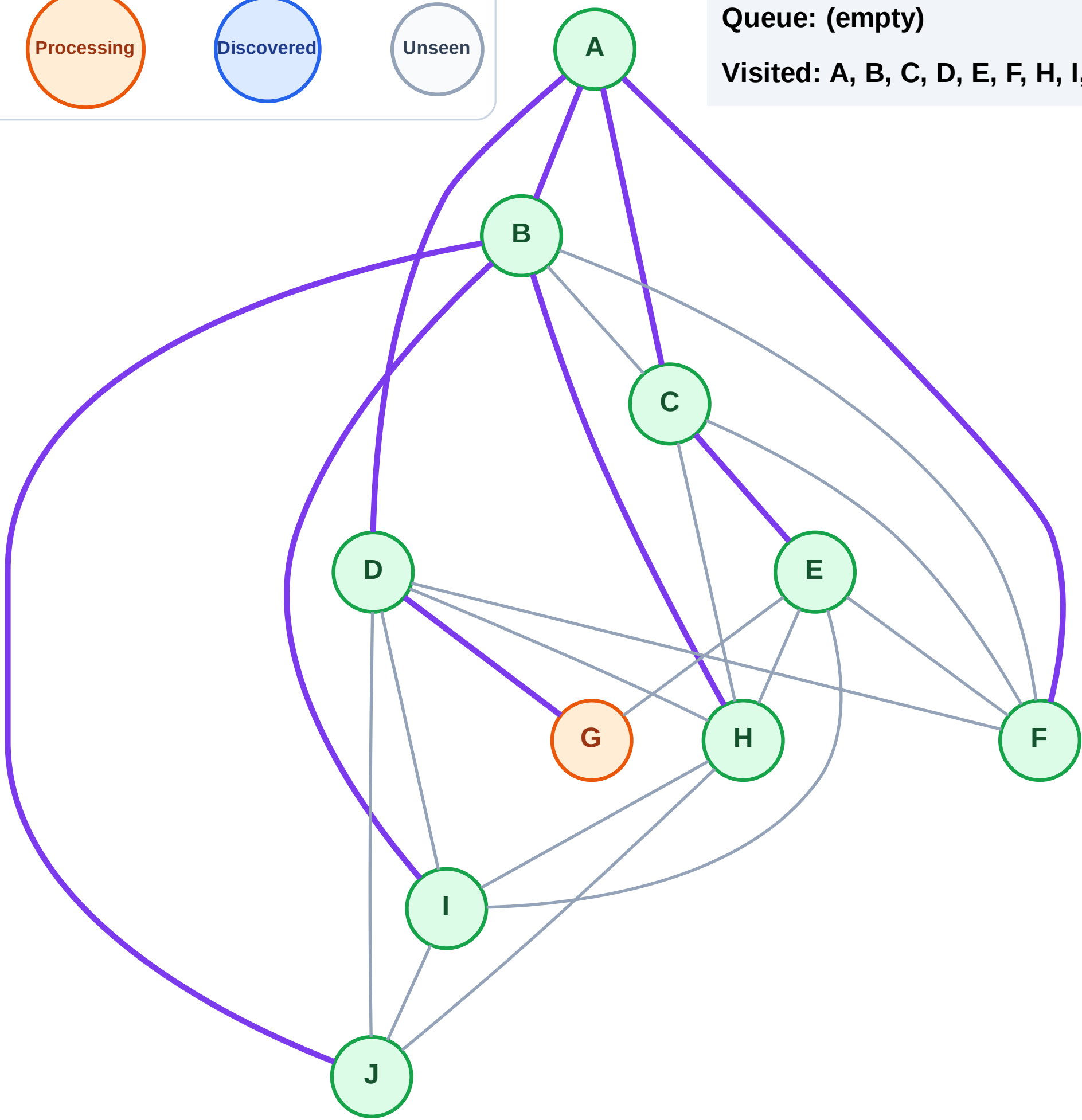
Step 20: Process node G

Legend



Queue: (empty)

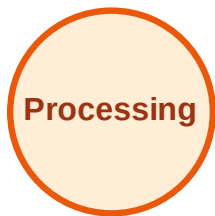
Visited: A, B, C, D, E, F, H, I, J



BFS Traversal

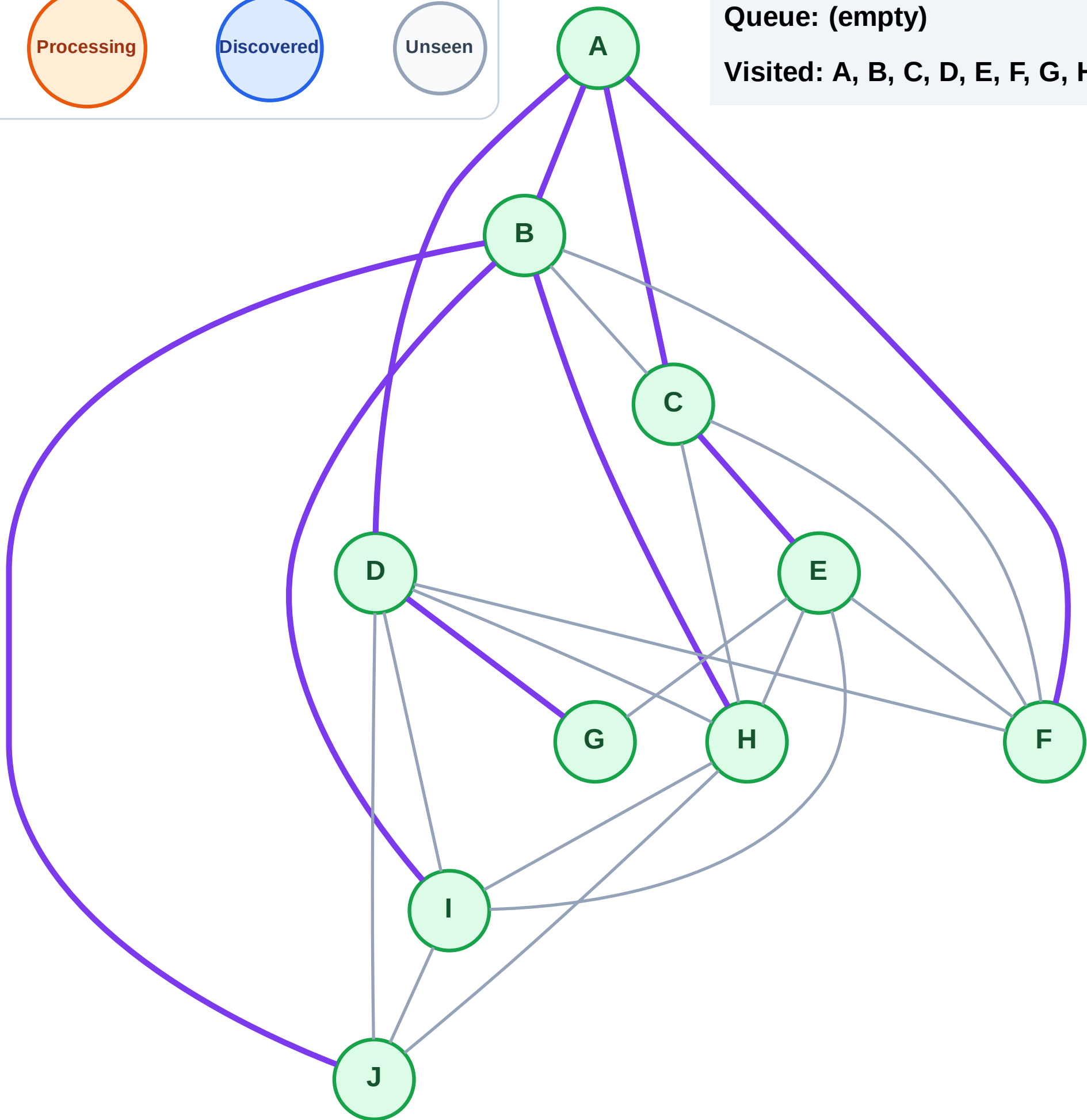
Step 21: No new neighbors from G

Legend



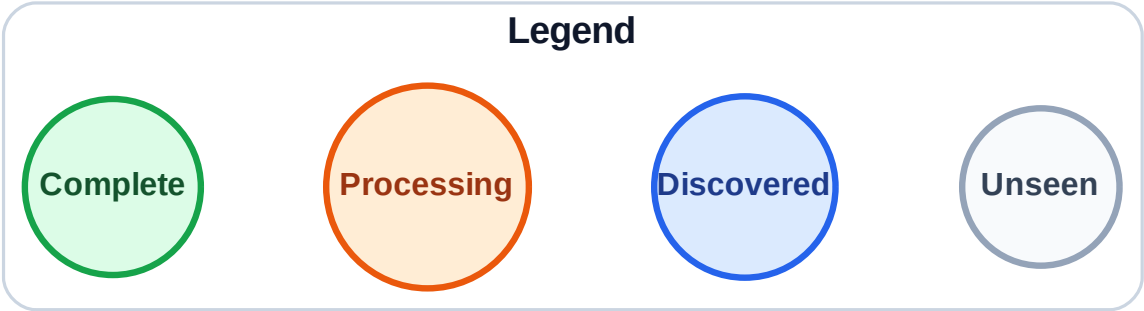
Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J

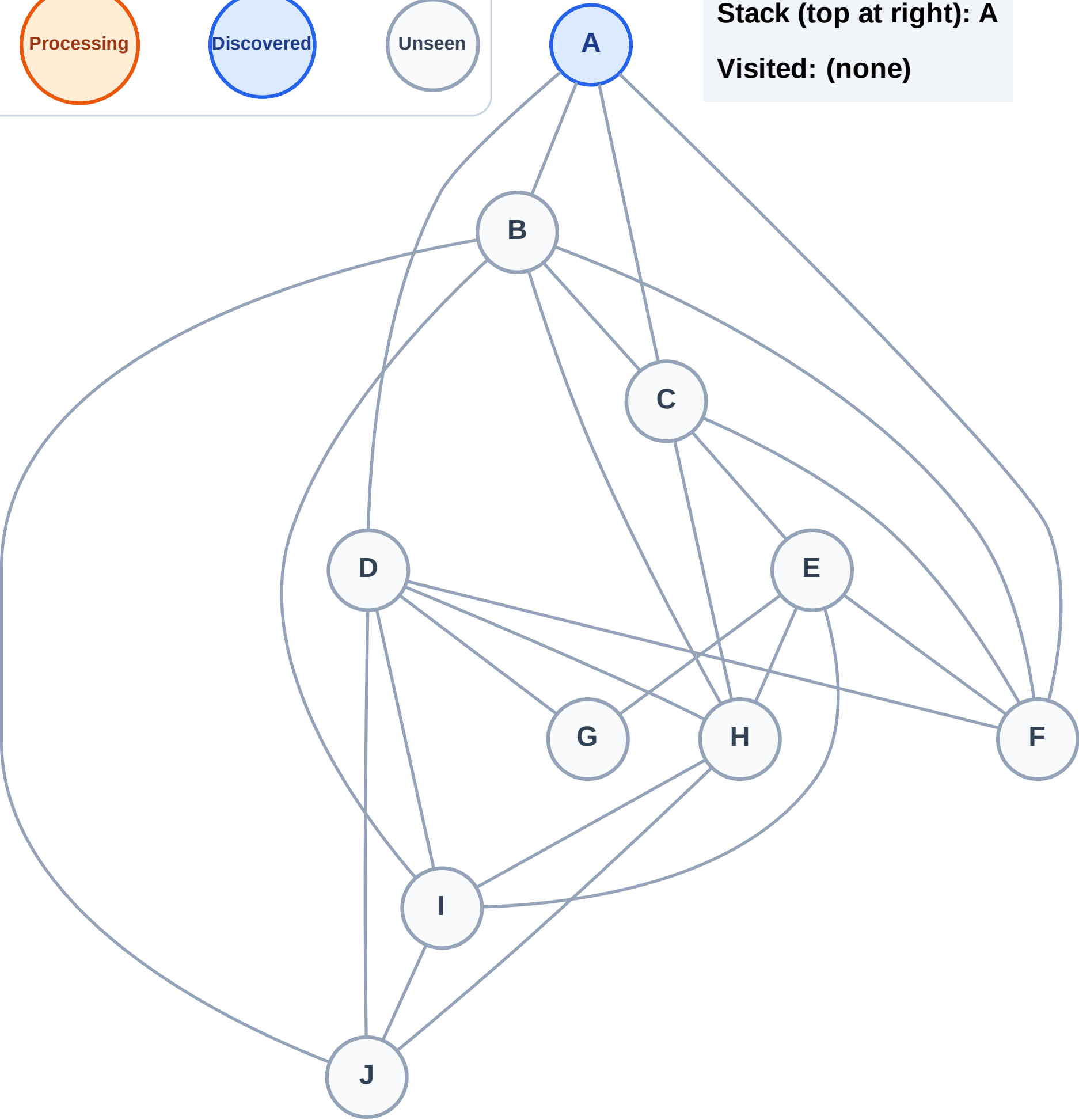


DFS Traversal

Step 1: Start at A



Stack (top at right): A
Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

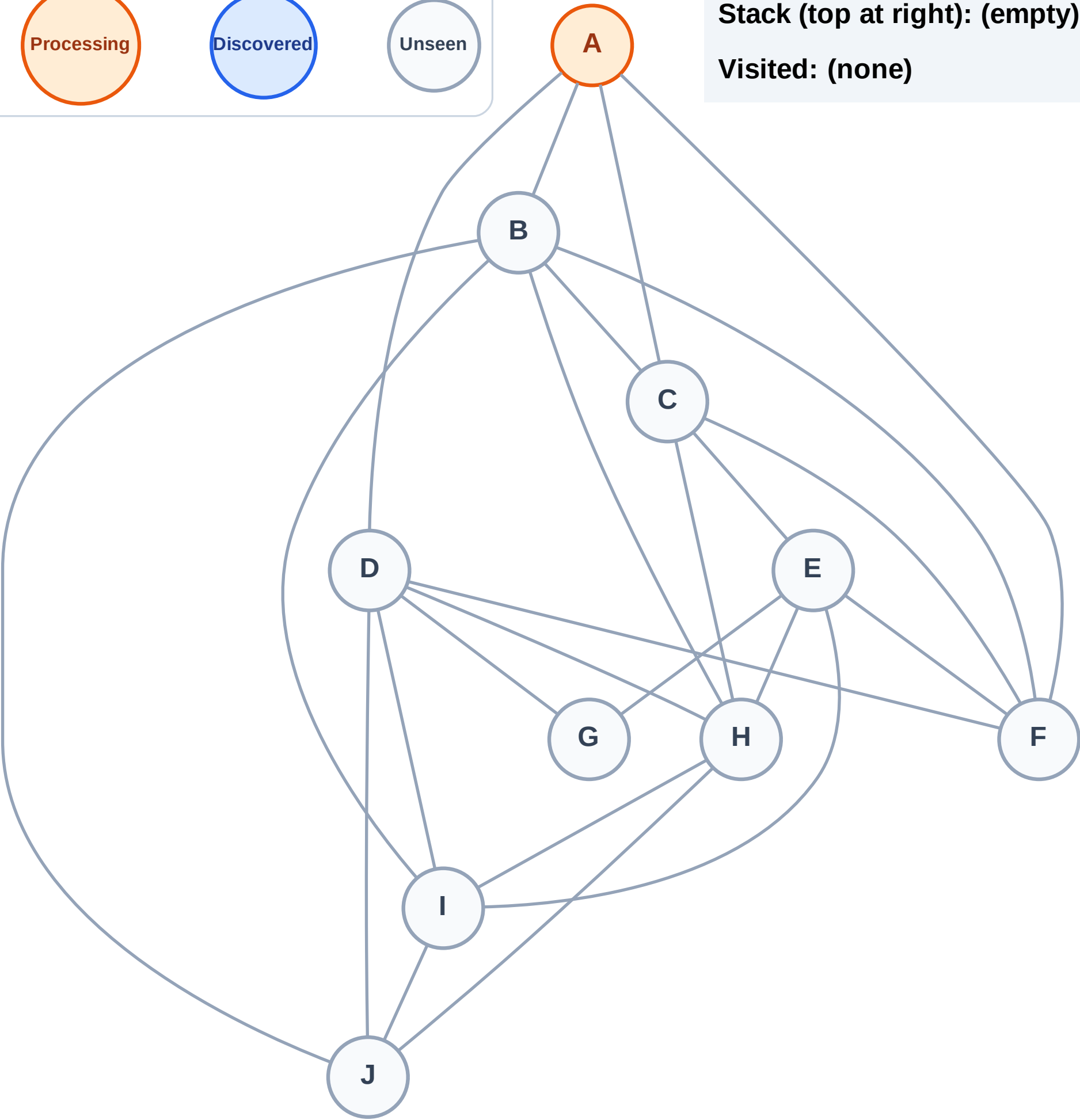
Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)
Visited: (none)



DFS Traversal

Step 3: Discover F, D, C, B from A

Legend

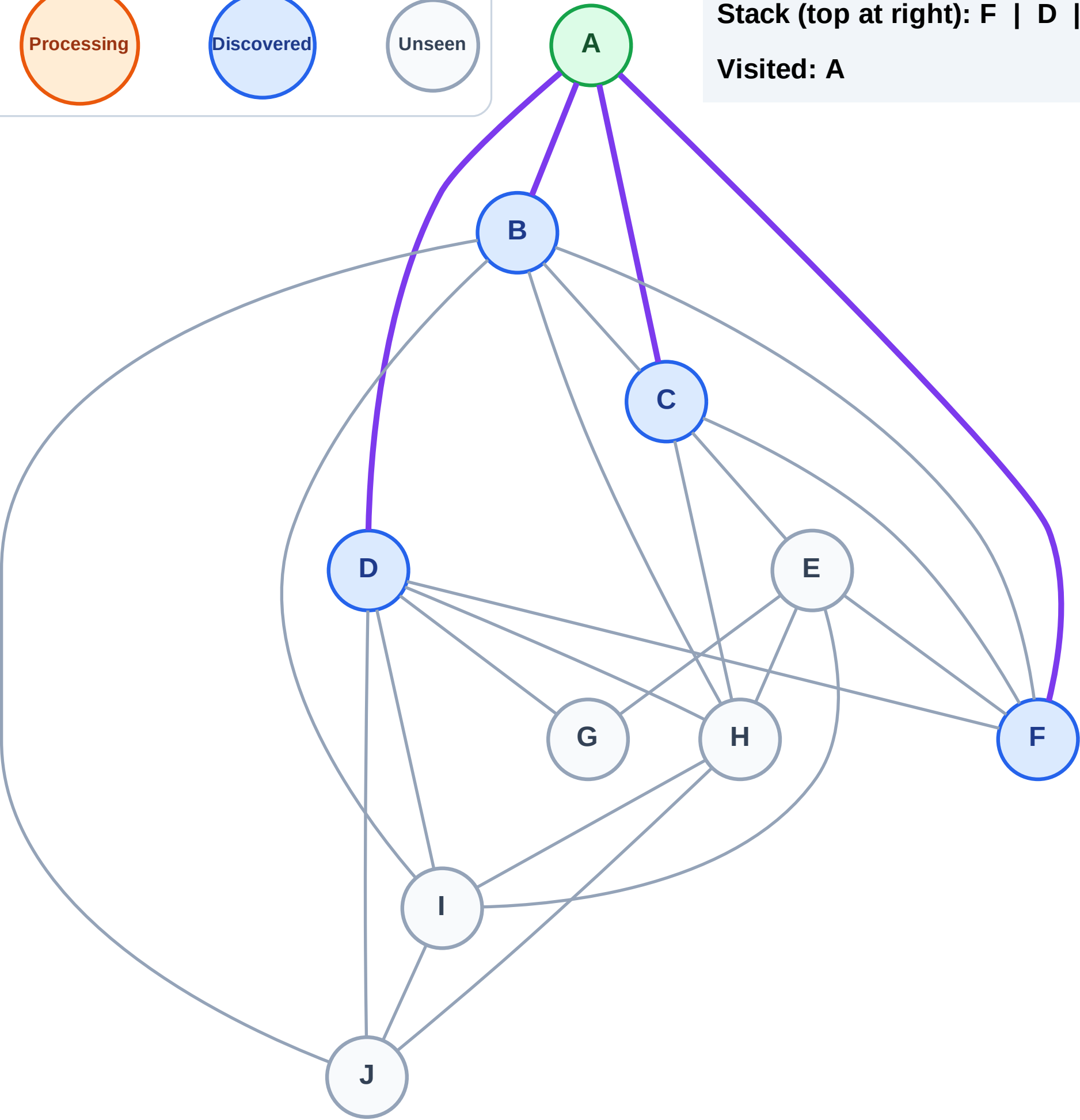
Complete

Processing

Discovered

Unseen

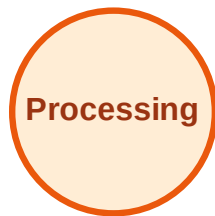
Stack (top at right): F | D | C | B
Visited: A



DFS Traversal

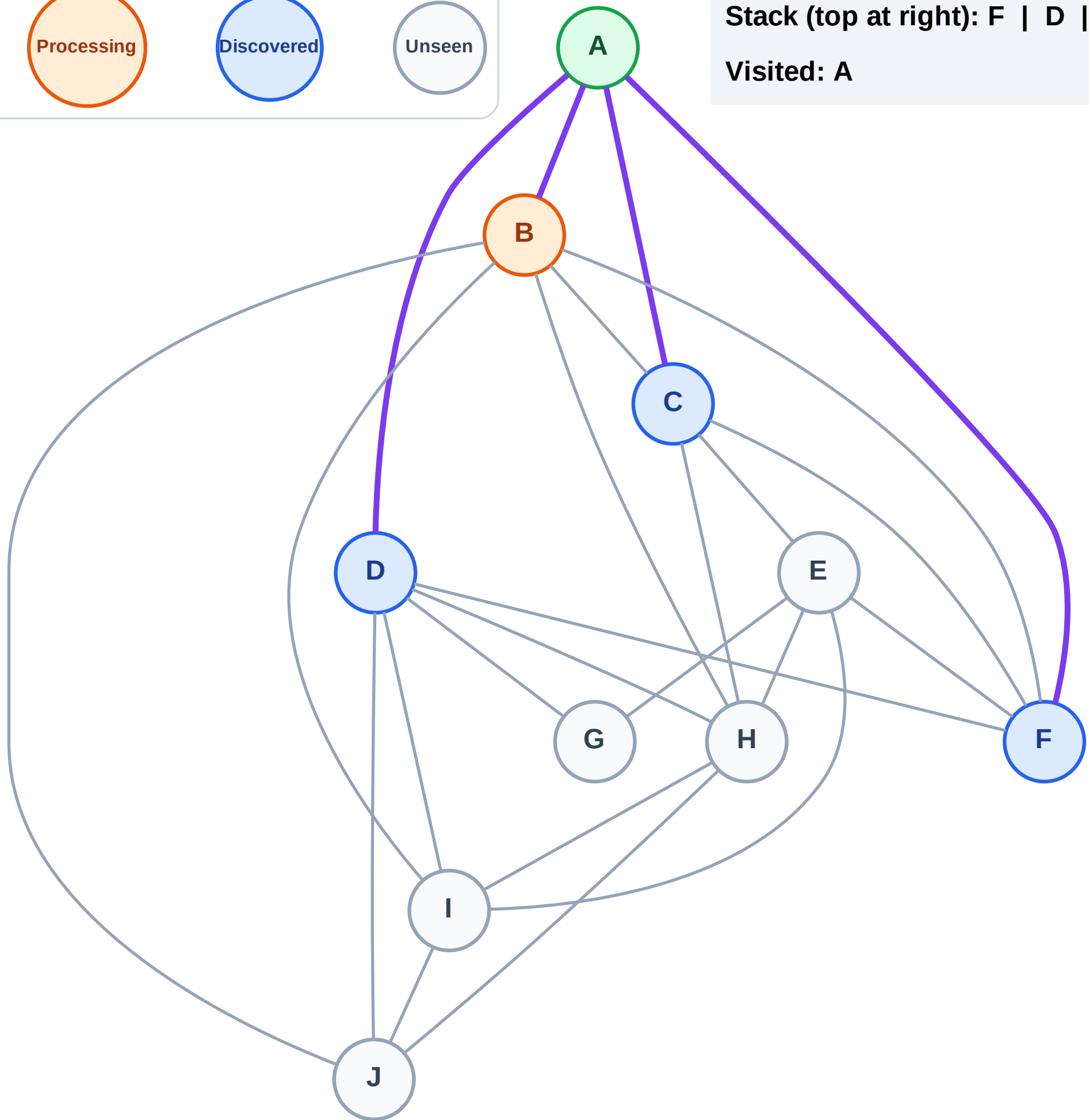
Step 4: Process node B

Legend



Stack (top at right): F | D | C

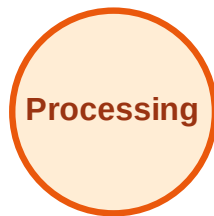
Visited: A



DFS Traversal

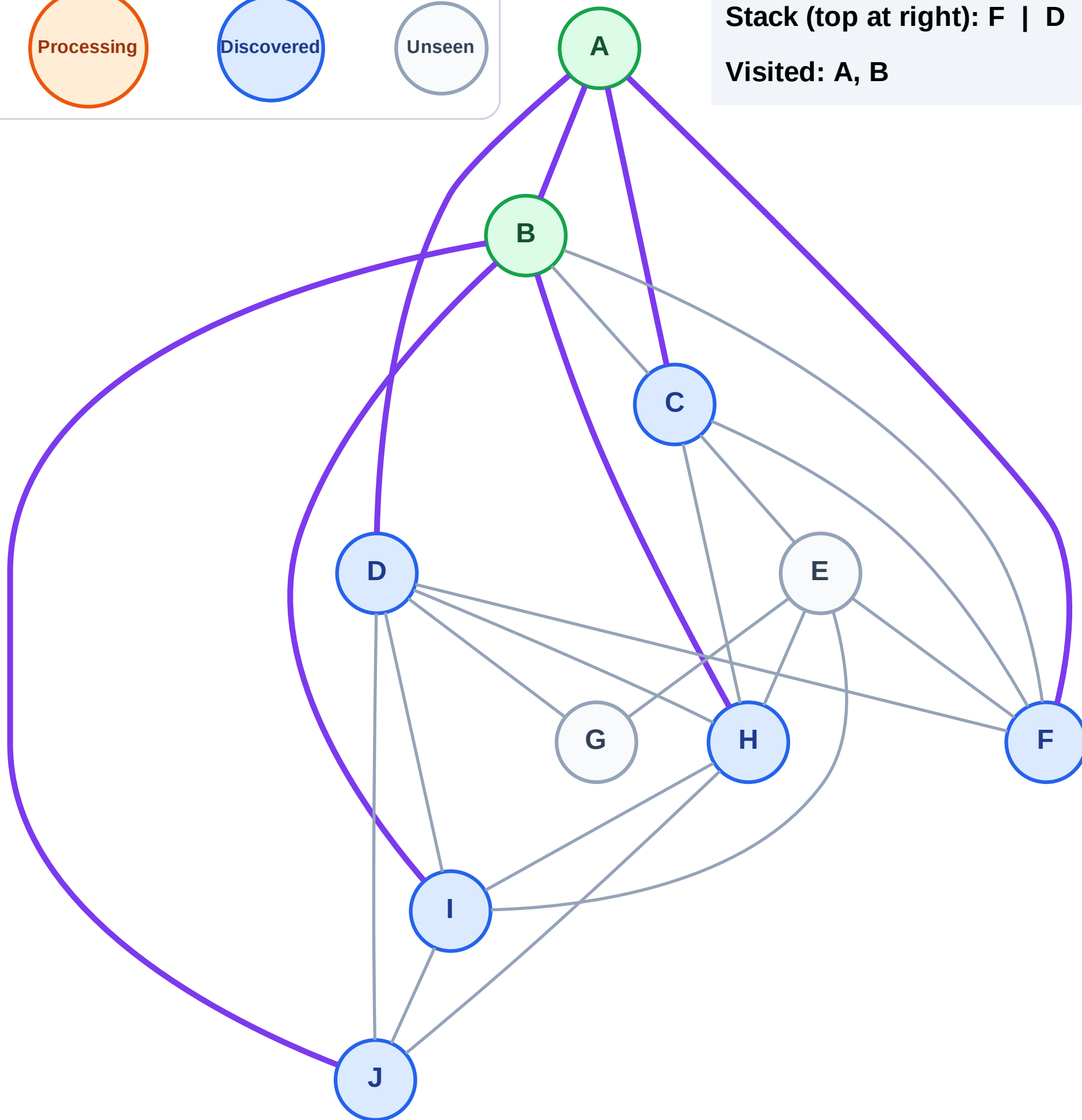
Step 5: Discover J, I, H from B

Legend



Stack (top at right): F | D | C | J | I | H

Visited: A, B



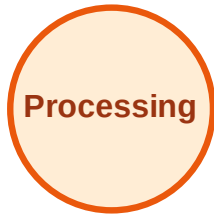
DFS Traversal

Step 6: Process node H

Legend



Complete



Processing



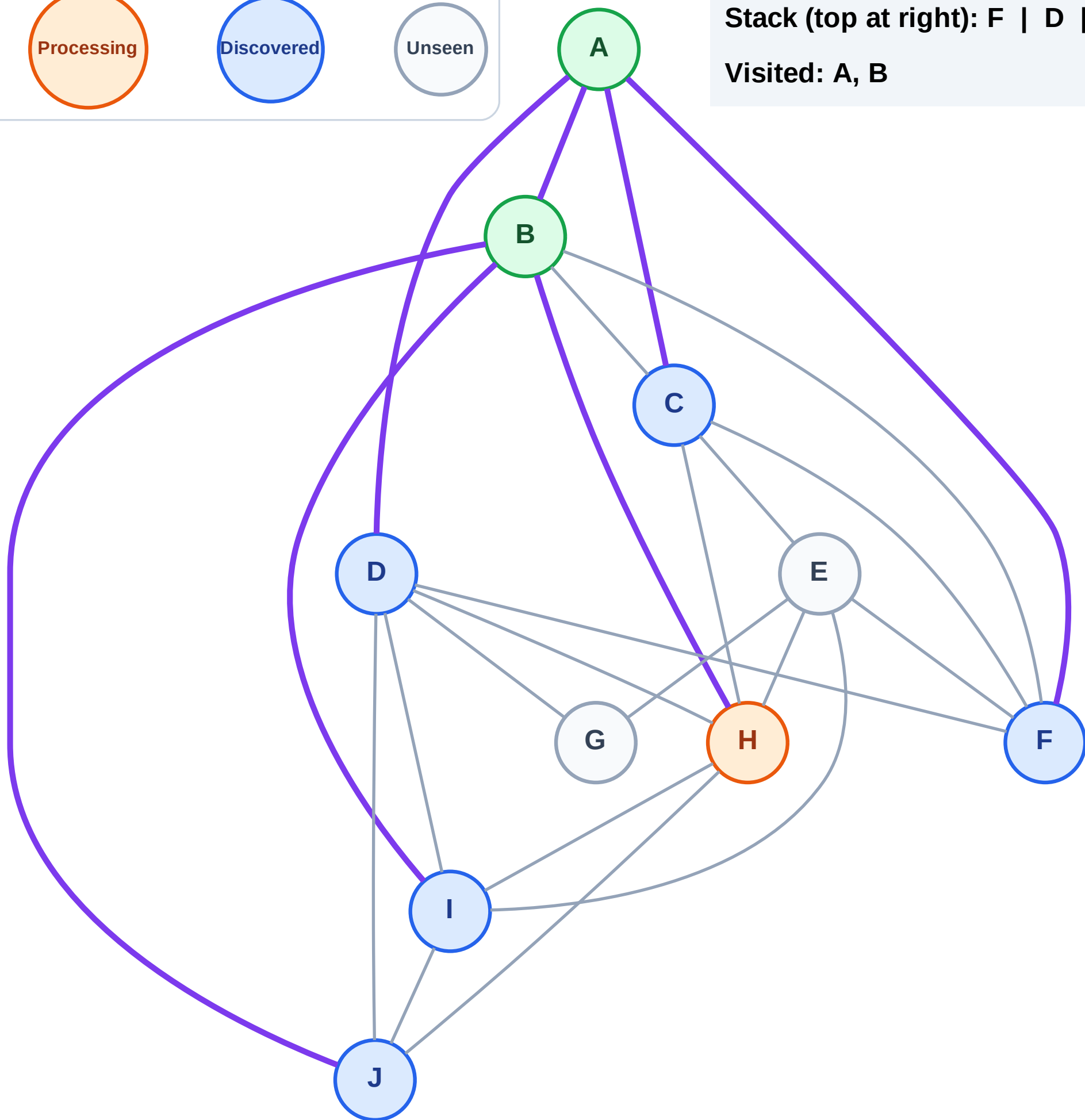
Discovered



Unseen

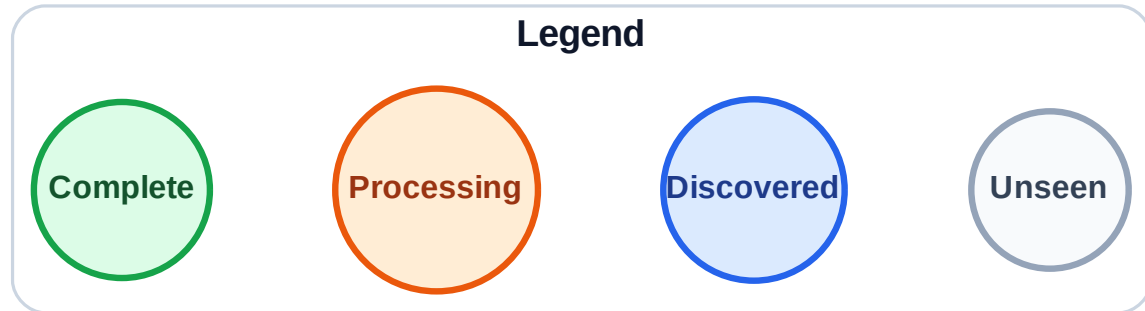
Stack (top at right): F | D | C | J | I

Visited: A, B



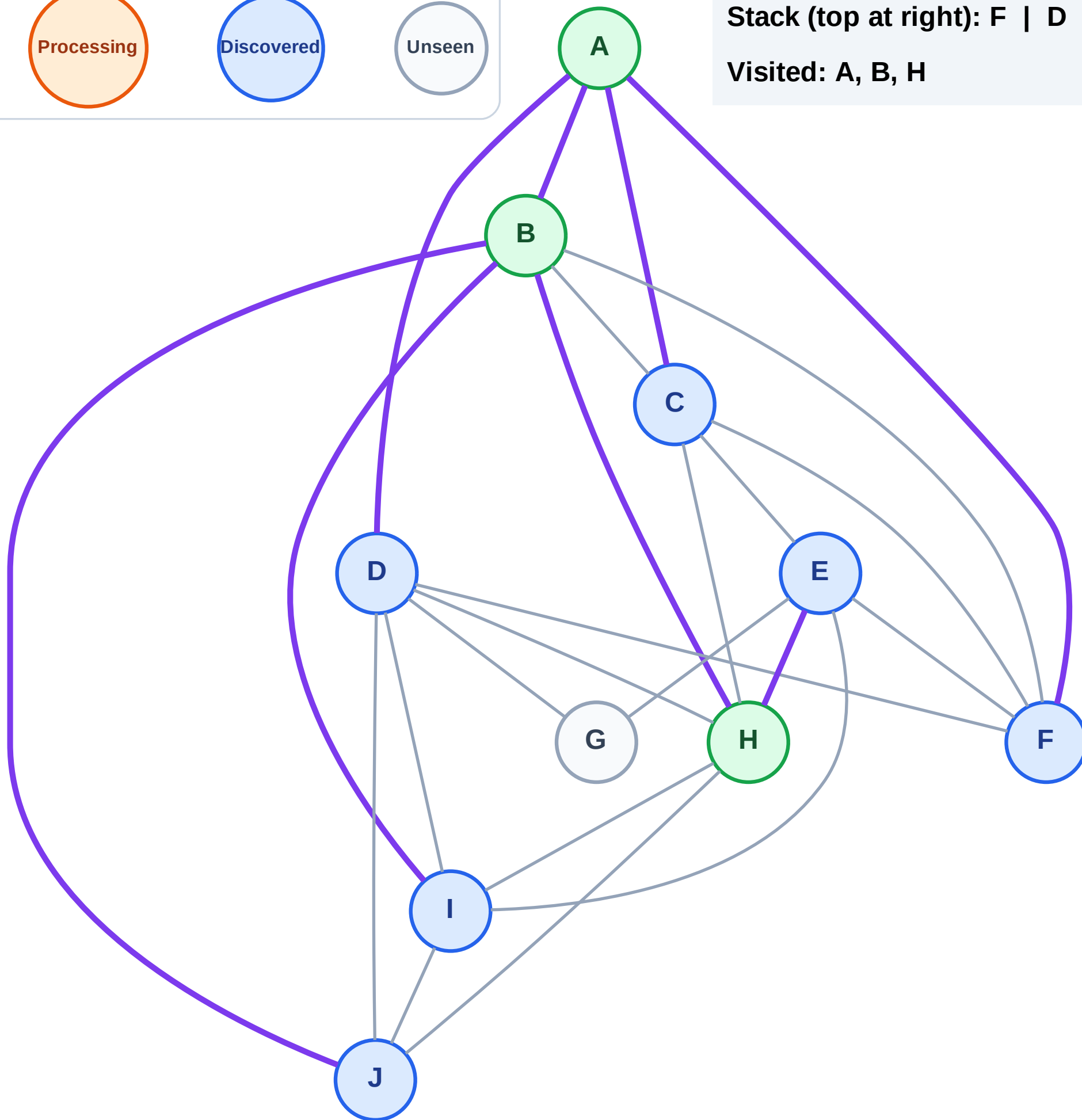
DFS Traversal

Step 7: Discover E from H



Stack (top at right): F | D | C | J | I | E

Visited: A, B, H



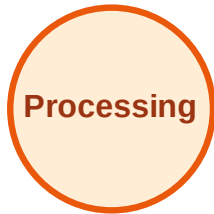
DFS Traversal

Step 8: Process node E

Legend



Complete



Processing



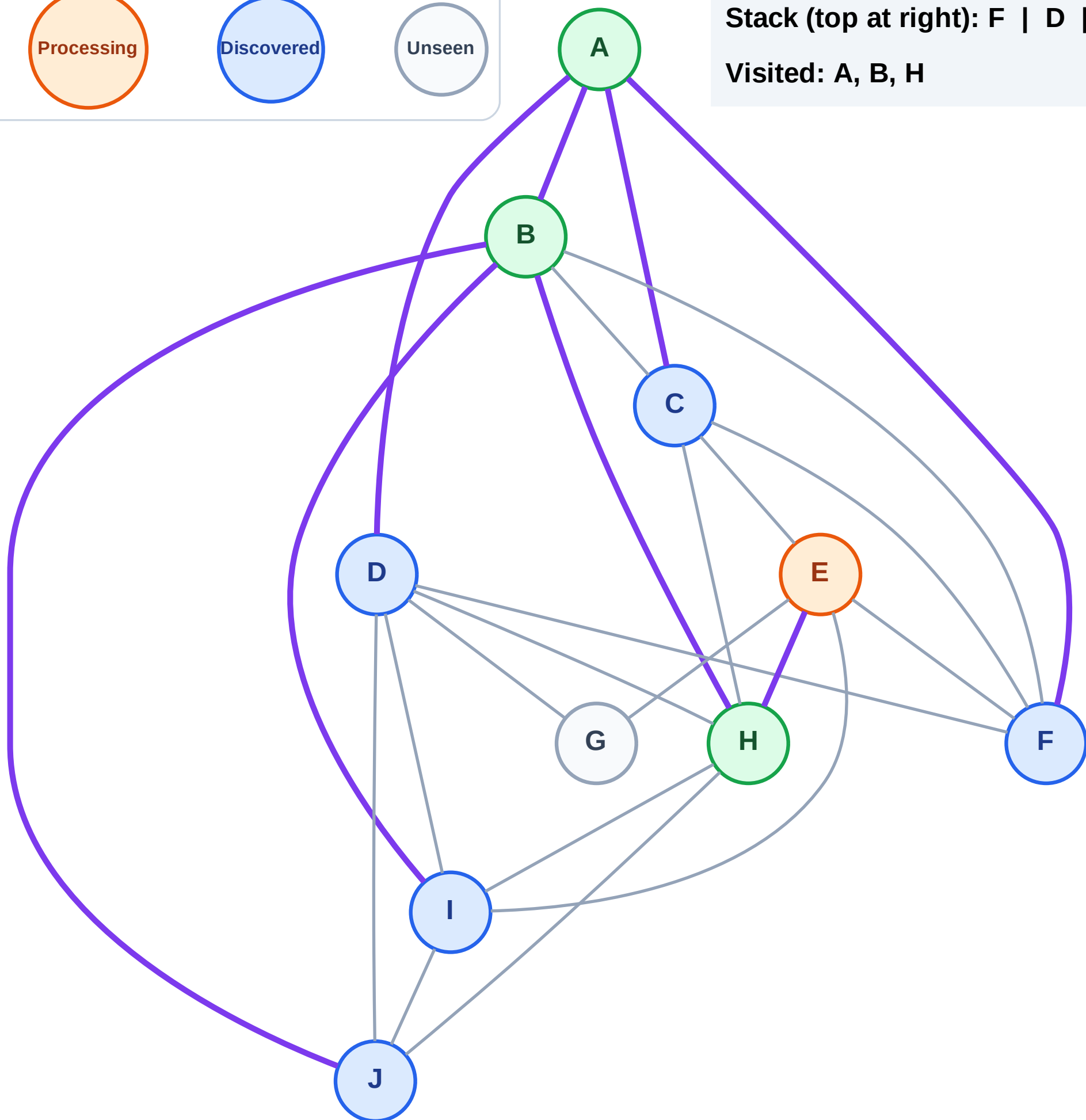
Discovered



Unseen

Stack (top at right): F | D | C | J | I

Visited: A, B, H



DFS Traversal

Step 9: Discover G from E

Legend

Complete

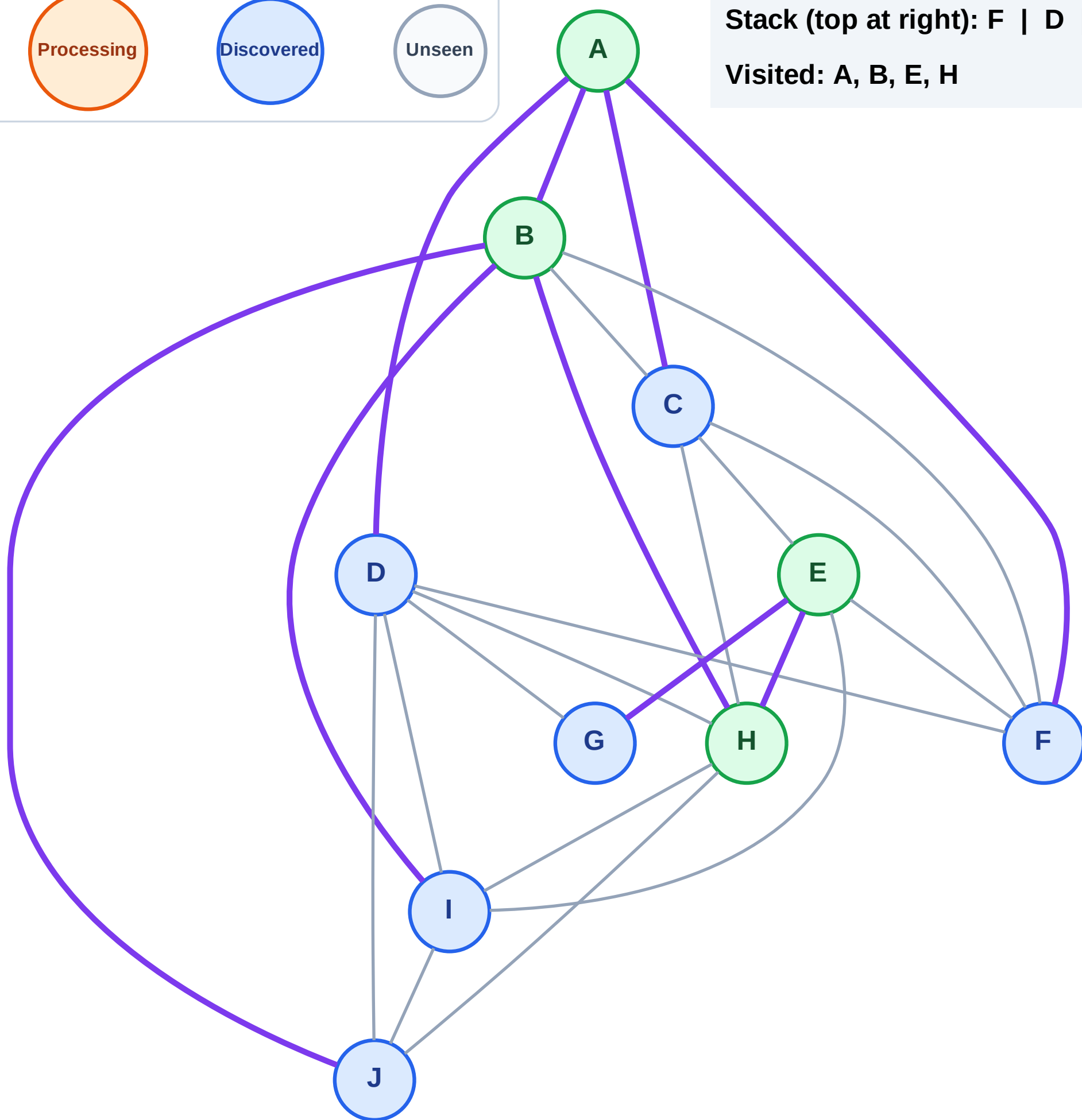
Processing

Discovered

Unseen

Stack (top at right): F | D | C | J | I | G

Visited: A, B, E, H



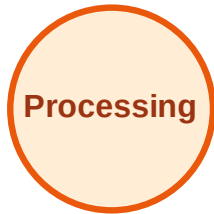
DFS Traversal

Step 10: Process node G

Legend



Complete



Processing



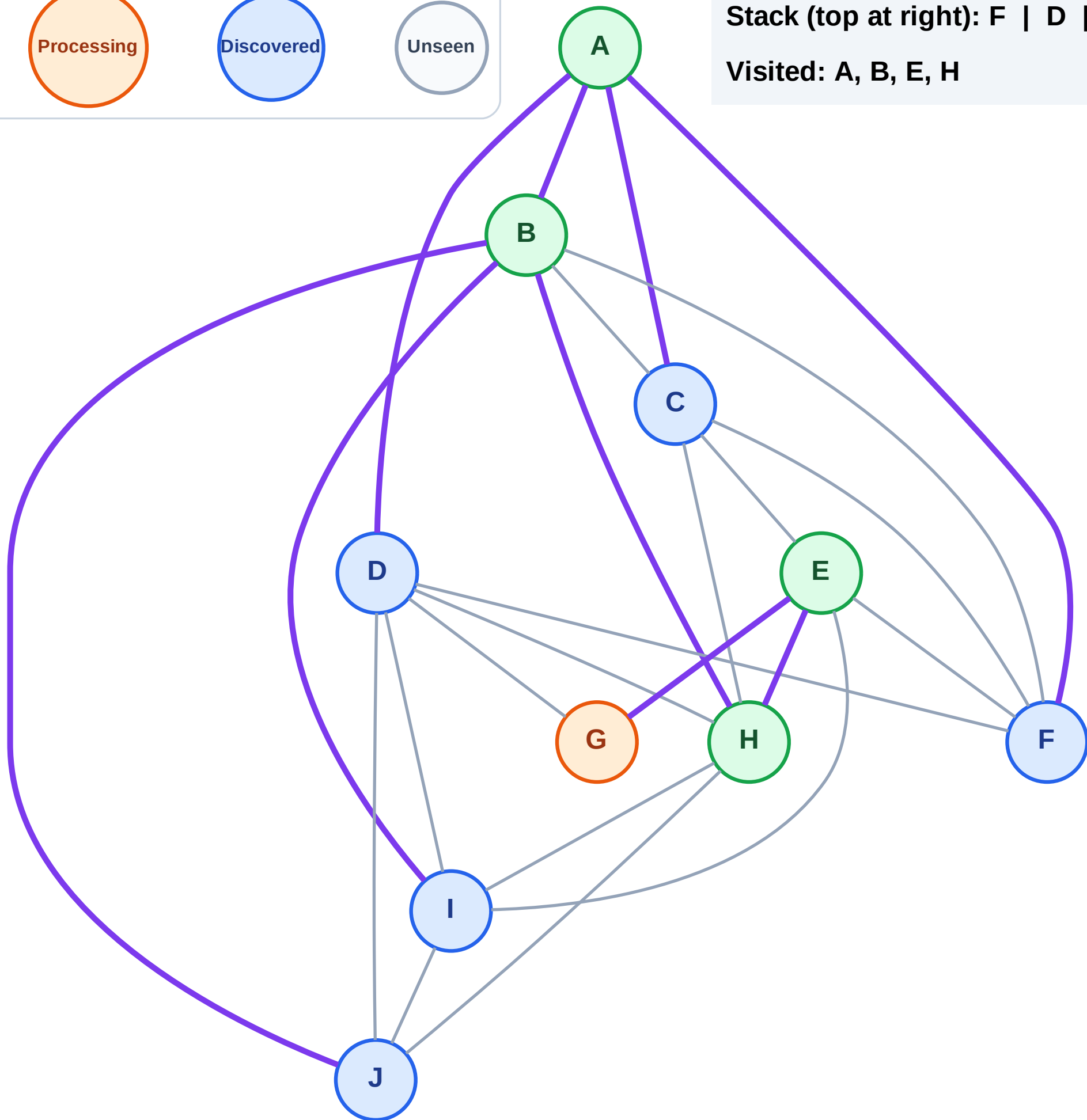
Discovered



Unseen

Stack (top at right): F | D | C | J | I

Visited: A, B, E, H



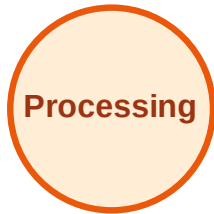
DFS Traversal

Step 11: No new neighbors from G

Legend



Complete



Processing



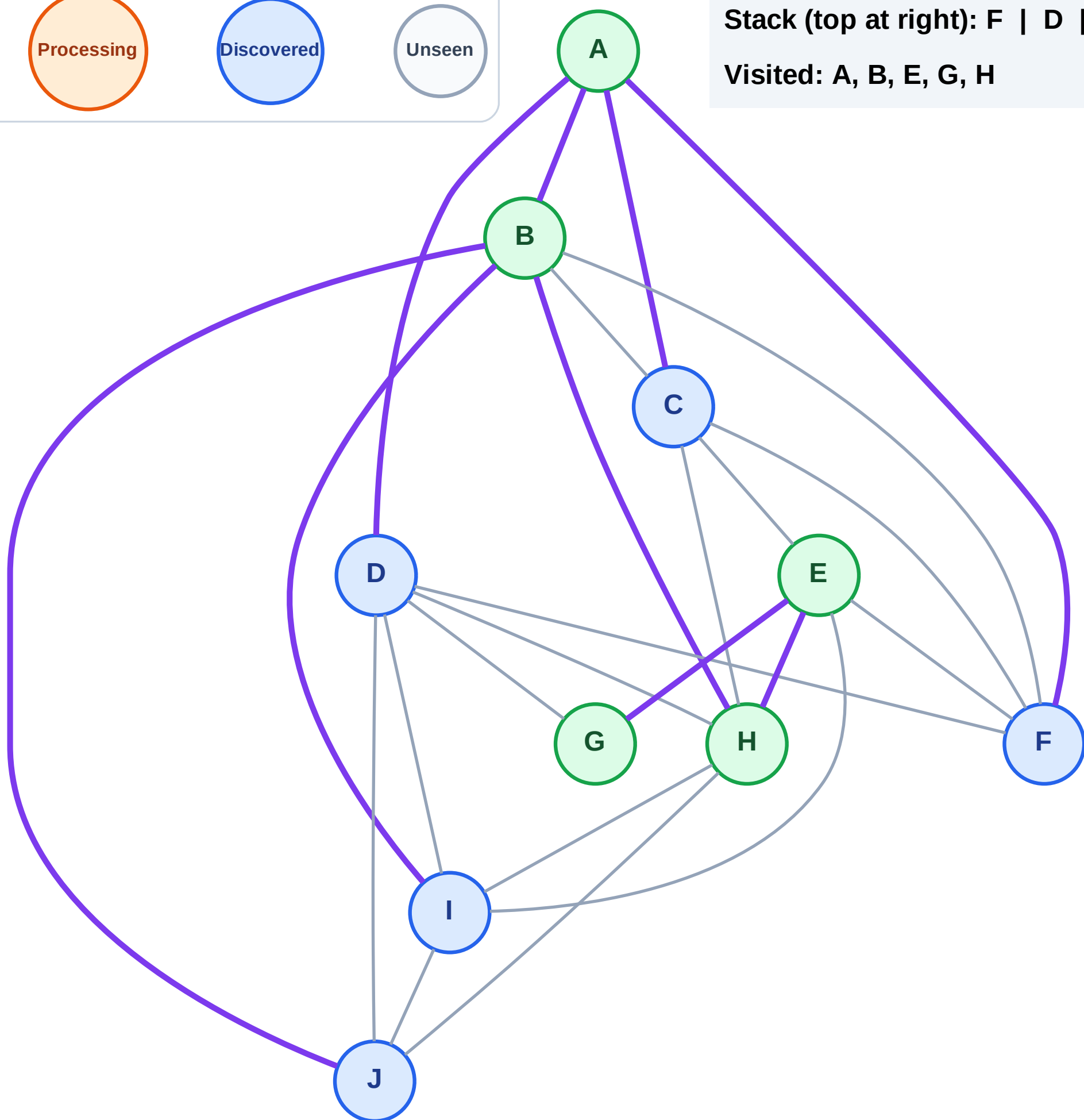
Discovered



Unseen

Stack (top at right): F | D | C | J | I

Visited: A, B, E, G, H



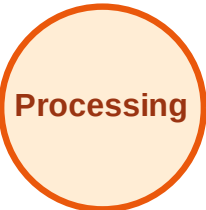
DFS Traversal

Step 12: Process node I

Legend



Complete



Processing



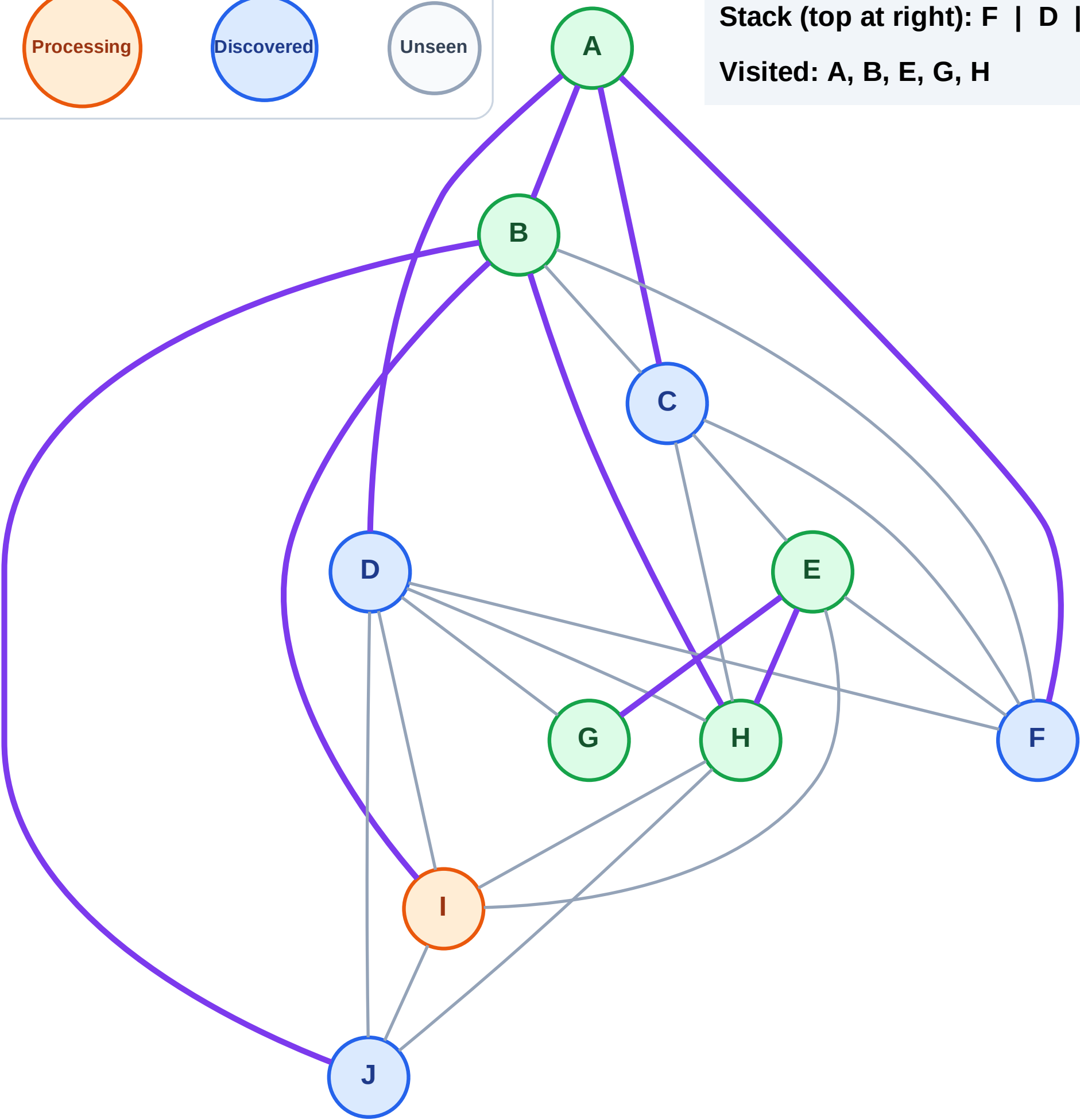
Discovered



Unseen

Stack (top at right): F | D | C | J

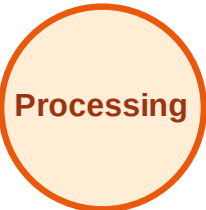
Visited: A, B, E, G, H



DFS Traversal

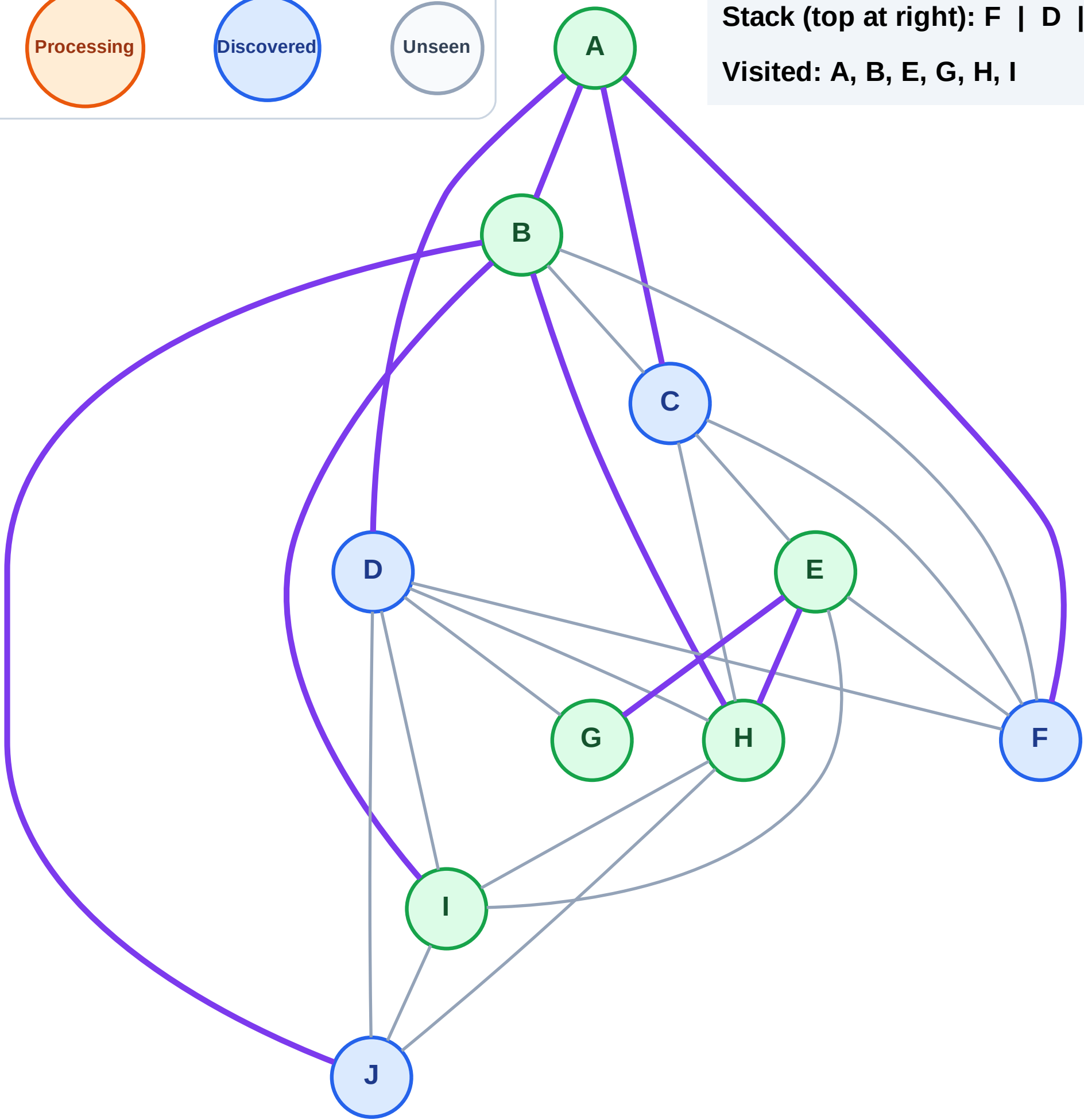
Step 13: No new neighbors from I

Legend



Stack (top at right): F | D | C | J

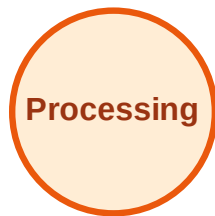
Visited: A, B, E, G, H, I



DFS Traversal

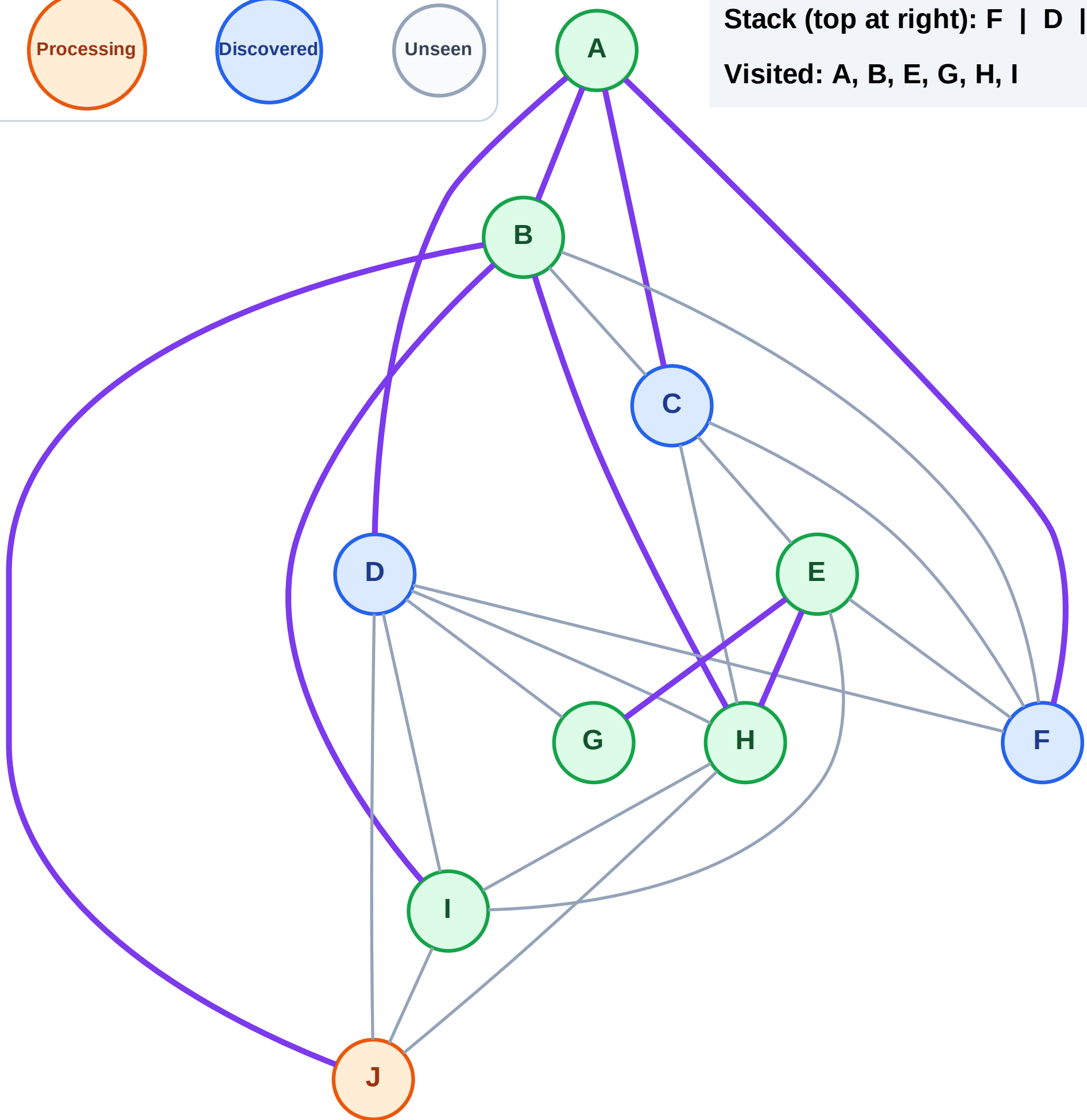
Step 14: Process node J

Legend



Stack (top at right): F | D | C

Visited: A, B, E, G, H, I



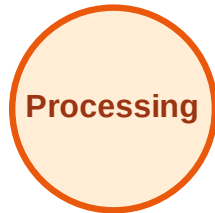
DFS Traversal

Step 15: No new neighbors from J

Legend



Complete



Processing



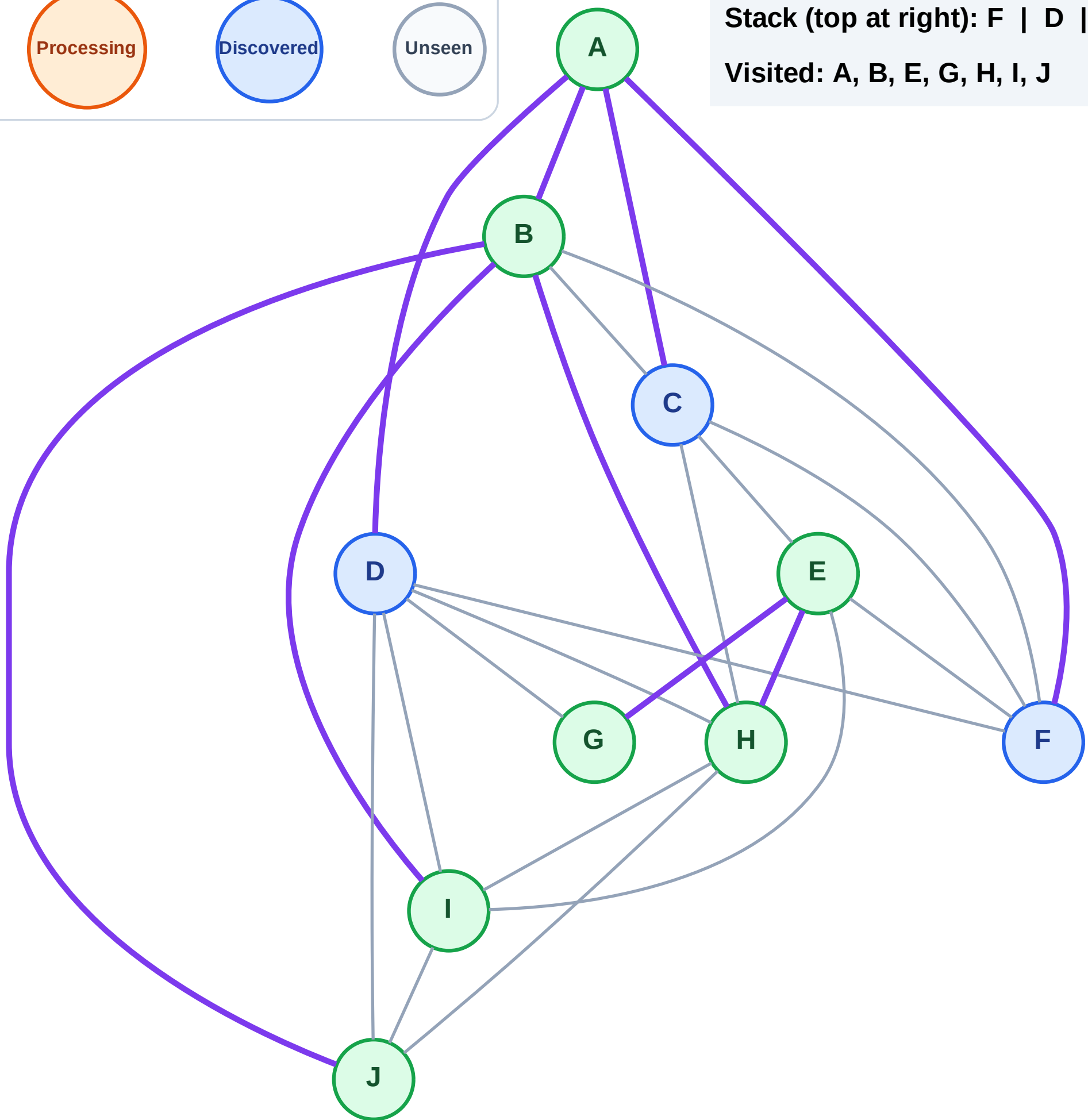
Discovered



Unseen

Stack (top at right): F | D | C

Visited: A, B, E, G, H, I, J



DFS Traversal

Step 16: Process node C

Legend

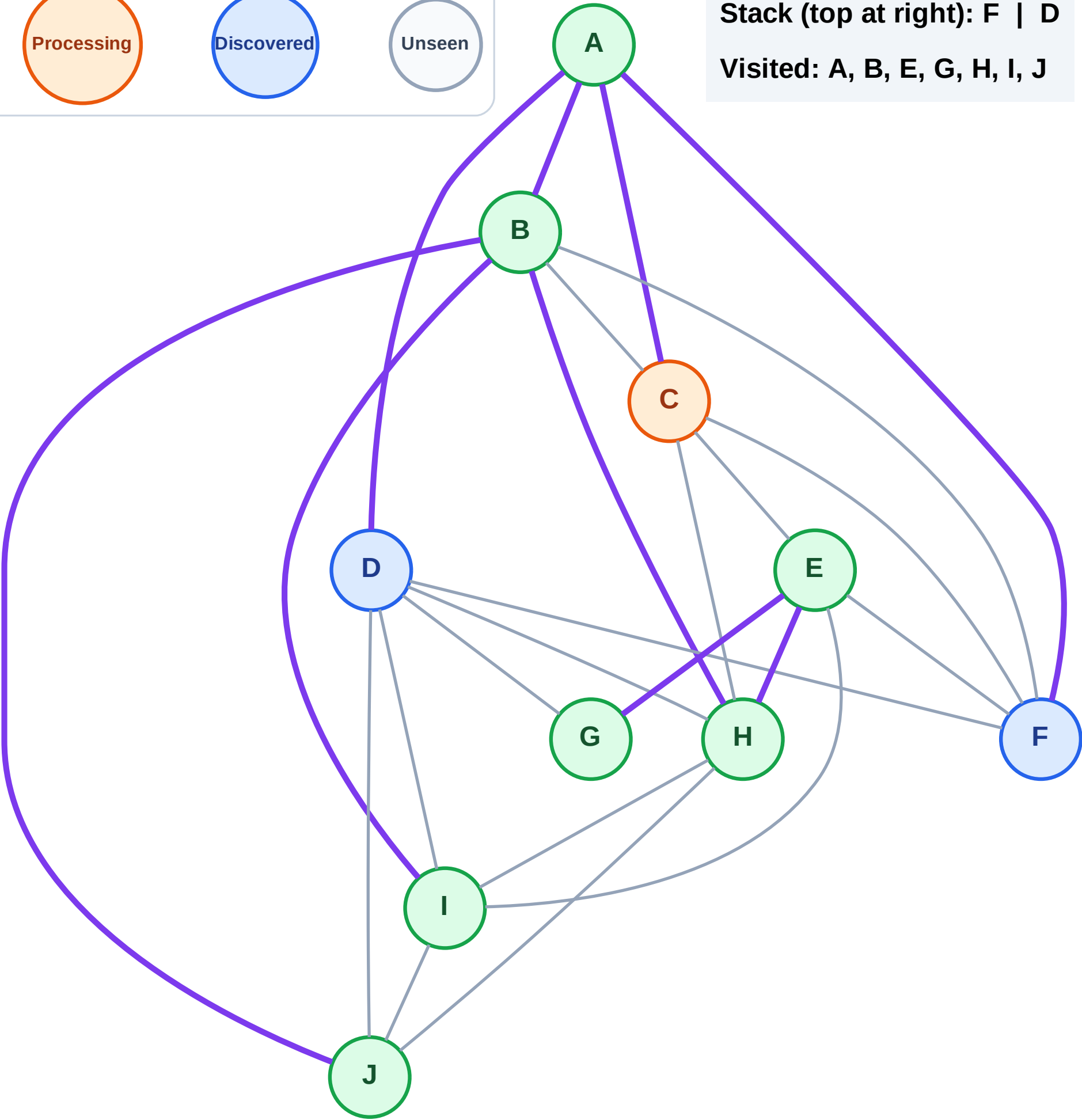
Complete

Processing

Discovered

Unseen

Stack (top at right): F | D
Visited: A, B, E, G, H, I, J



DFS Traversal

Step 17: No new neighbors from C

Legend

Complete

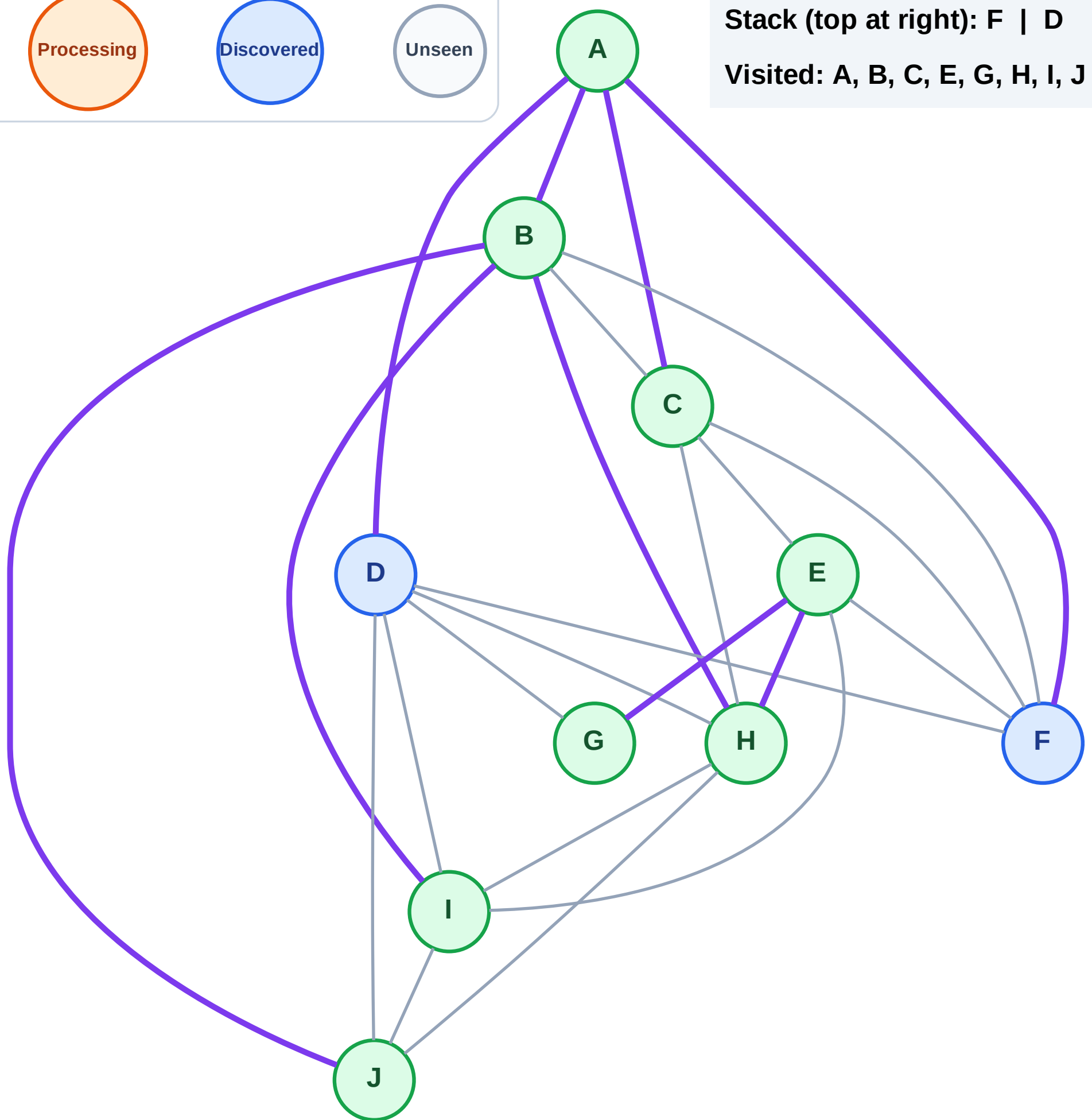
Processing

Discovered

Unseen

Stack (top at right): F | D

Visited: A, B, C, E, G, H, I, J



DFS Traversal

Step 18: Process node D

Legend

Complete

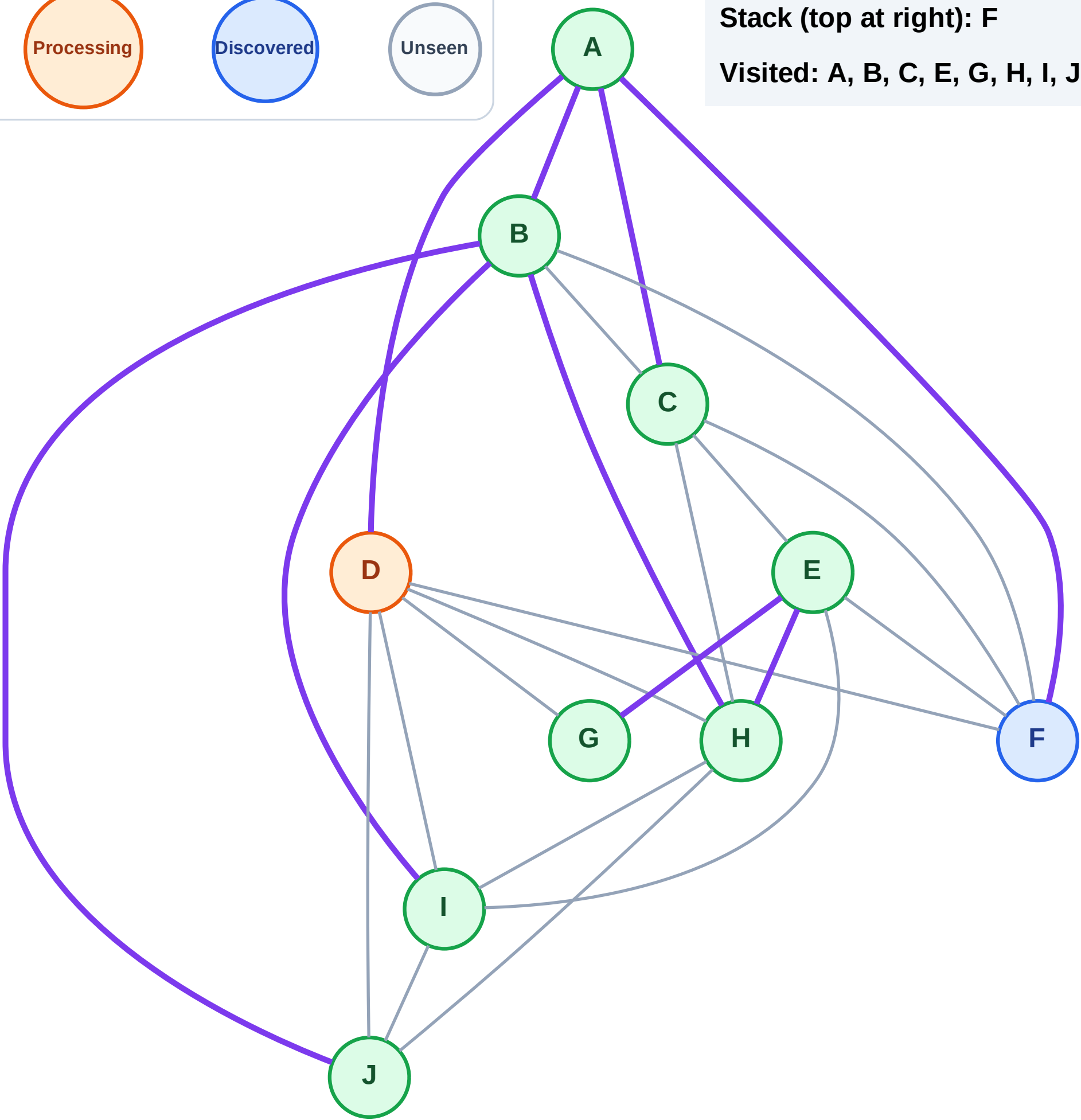
Processing

Discovered

Unseen

Stack (top at right): F

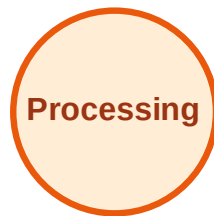
Visited: A, B, C, E, G, H, I, J



DFS Traversal

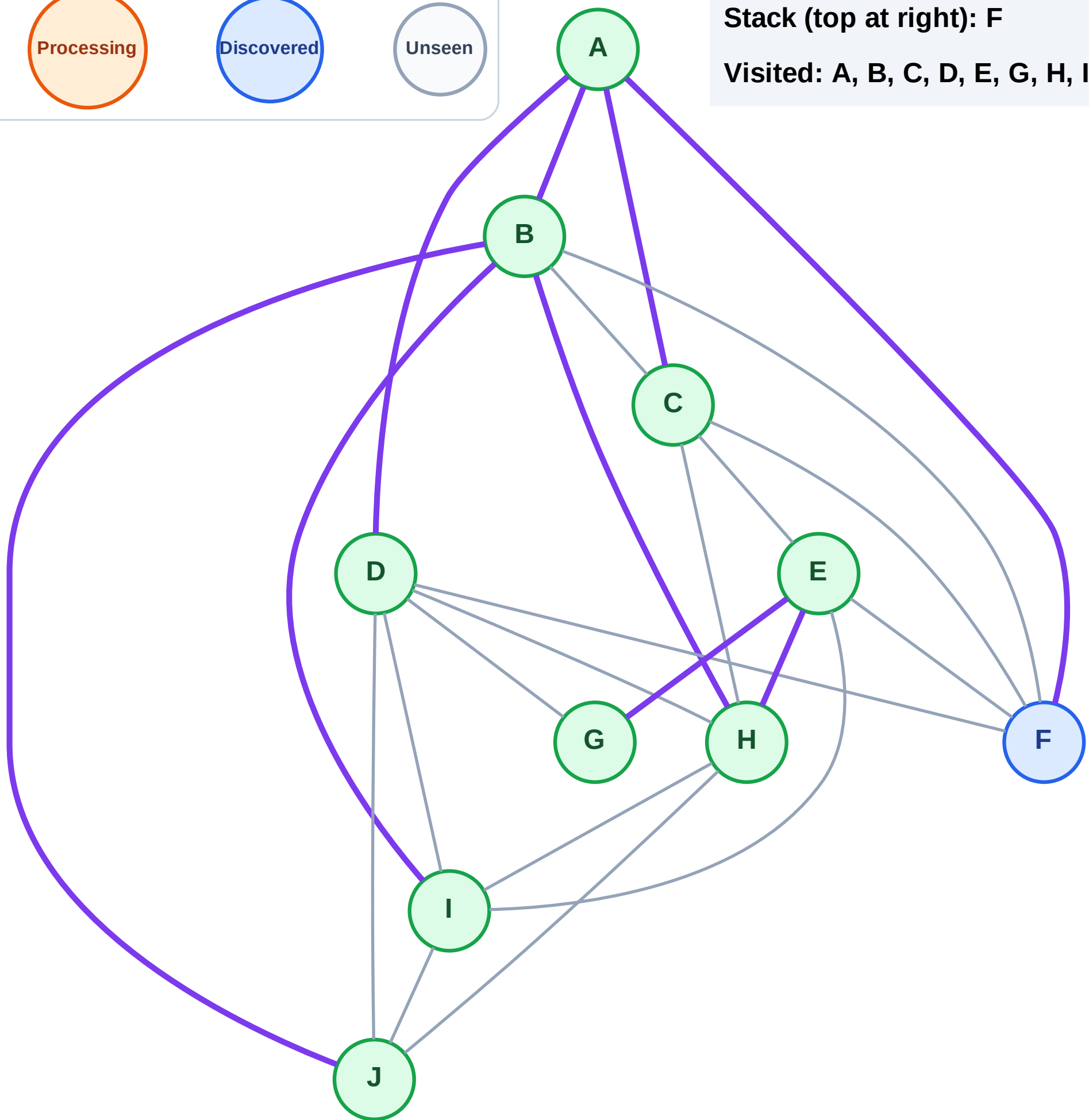
Step 19: No new neighbors from D

Legend



Stack (top at right): F

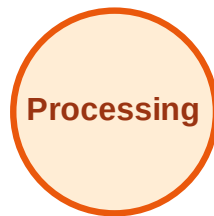
Visited: A, B, C, D, E, G, H, I, J



DFS Traversal

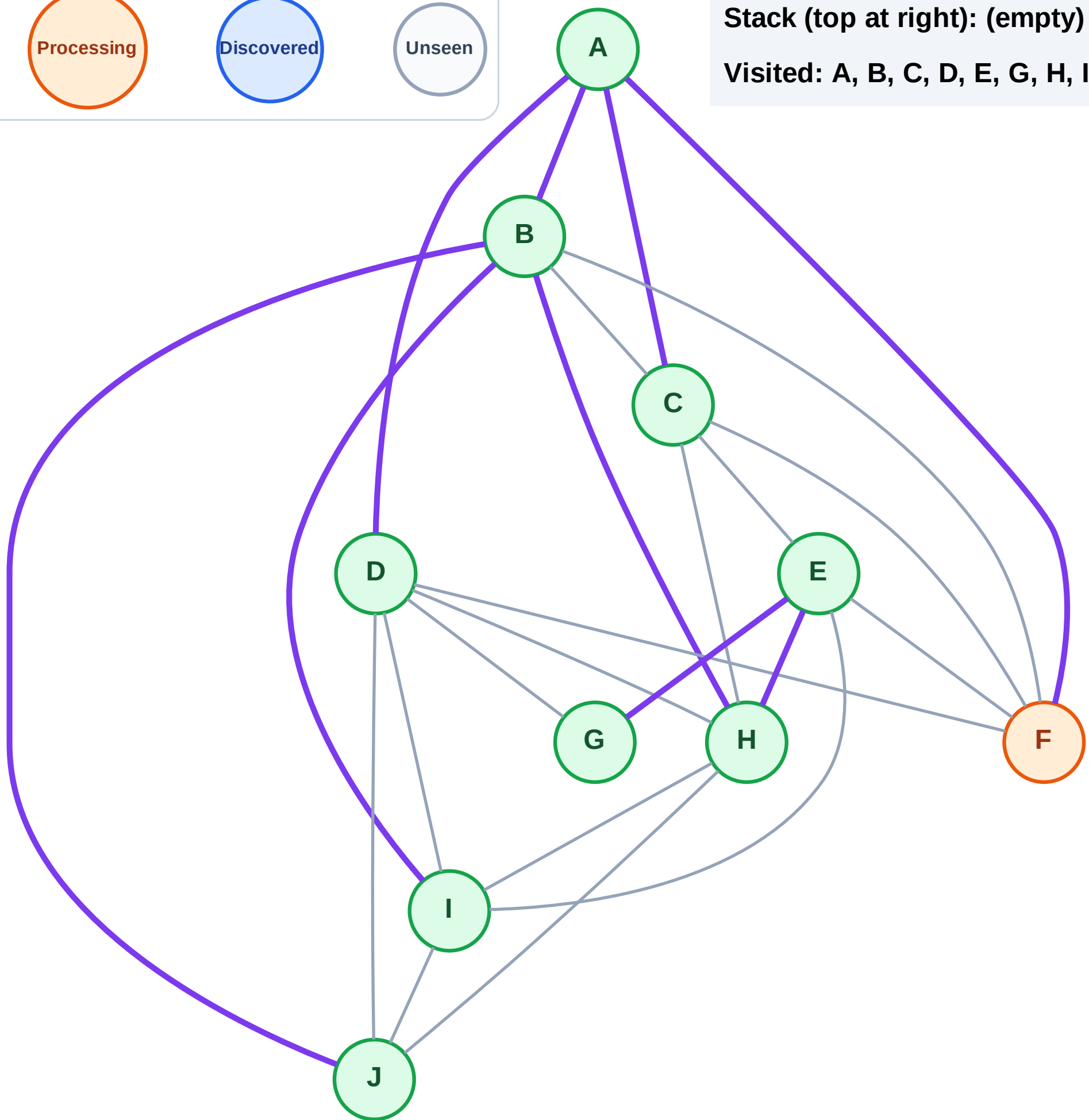
Step 20: Process node F

Legend



Stack (top at right): (empty)

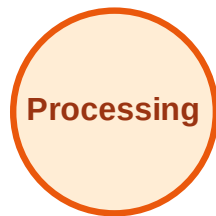
Visited: A, B, C, D, E, G, H, I, J



DFS Traversal

Step 21: No new neighbors from F

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

